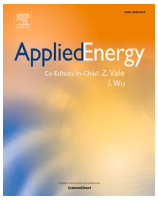
Applied Energy 421 (2026) 128106



Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apen





Wind power forecasting with deep representation: A survey

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HIGHLIGHTS

- Reviews recent advances in large models, graph neural and recurrent neural methods.
- Summarizes transfer learning and federated learning for cross-scenario forecasting.
- Systematically reviews data preprocessing methods for wind power forecasting.
- Provides application-oriented analysis from scenario and value perspectives.
- Summarizes experimental settings, forecasting scenarios, horizons, and results.

ARTICLE INFO

Keywords:
Wind power forecasting
Deep representation models
Attention mechanism
Data decomposition
Application analysis
Learning strategy

ABSTRACT

Accurate wind power forecasting (WPF) is essential for ensuring the secure operation of power systems and supporting the sustainable development of renewable energy. Against the backdrop of rapid renewable energy development, achieving high-accuracy forecasting under complex meteorological conditions and diverse application scenarios remains a central objective in this field. In recent years, deep representation learning models have become a research focus in WPF owing to their strengths in nonlinear modeling and the representation of complex data structures. This review provides a systematic overview of current progress in deep representation models, with particular emphasis on pre-trained large models, graph neural networks, and recurrent neural networks, and further summarizes the roles of attention mechanisms across different dimensions. Typical issues and solutions in data preprocessing are outlined, and various signal decomposition methods are comparatively analyzed. In addition, publicly available datasets are presented, experimental details of representative methods are synthesized, and challenges across diverse application scenarios are addressed. Building on current trends, this work proposes future research directions, aiming to provide a structured reference for subsequent studies in WPF.

1. Introduction

As a major form of renewable and clean energy, wind energy has attracted increasing global attention in recent years due to its inexhaustible availability, environmental friendliness, and sustainable utilization. In response to the escalating challenges of climate change and the carbon reduction and neutrality targets proposed by many countries, wind energy demonstrates unique advantages in mitigating greenhouse gas emissions and reducing dependence on non-renewable resources. It is also becoming a key pillar in driving energy transition and achieving green and low-carbon development. According to IRENA's Renewable Capacity Statistics 2025 report [1], global installed wind power capacity reached 1132.66 GW by the end of 2024, representing a year-on-year growth of 11.5%. Over the past two decades, global installed wind power capacity has maintained rapid growth (as illustrated in Fig. 1), and wind energy has gradually become an integral part of smart grids, regional microgrids, and distributed energy systems [2].

However, the inherent randomness, volatility, and intermittency of wind energy make its power output difficult to control [3]. This not only poses severe challenges to the safety and stability of large-scale wind power integration, but also increases uncertainties in power system scheduling, load balancing, and electricity market trading. Therefore,

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https://doi.org/10.1016/j.apenergy.2026.128106
Received 10 February 2026; Received in revised form 16 May 2026; Accepted 24 May 2026
Available online 18 June 2026
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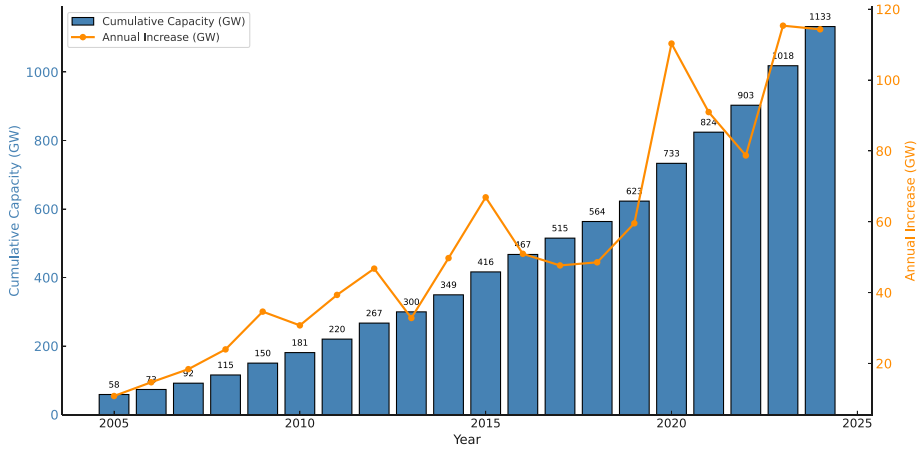


Fig. 1. Newly installed capacity and cumulative installed capacity of wind power worldwide from 2005 to 2024.

accurate and reliable wind power forecasting (WPF) technologies are gradually becoming a critical foundation for ensuring the secure operation of power systems, enhancing wind power integration, and promoting the development of renewable energy [4].

Currently, WPF methods can be broadly categorized into three groups: physical models, statistical models, and machine learning models. Physical models generally rely on meteorological data and turbine parameters, using mathematical equations to estimate power output. These methods depend less on historical data, which provides advantages for newly built wind farms. However, they are computationally intensive and require substantial resources [5]. Statistical models are primarily based on time series analysis and statistical principles, leveraging historical observational data to capture temporal correlations [6]. Although these methods demonstrated significant effectiveness in early applications, they rely on large and stable datasets. Their ability to characterize non-stationary and nonlinear features, such as wind speed and wind direction, is limited, making it difficult to maintain accuracy under complex meteorological conditions [7]. Machine learning methods include both classical machine learning and deep learning (DL) approaches. Classical machine learning methods, such as support vector regression [8] and decision trees [9], have shown strong performance in nonlinear forecasting tasks and are widely used in WPF. However, these classical approaches generally rely on manual feature engineering [10]. In contrast, DL models, built upon multi-layer neural architectures, can automatically learn hierarchical feature representations and better capture complex temporal, spatial, and nonlinear dependencies [11]. Owing to this superior representational capacity and their ability to extract multi-source spatiotemporal features, DL methods offer unique advantages in improving forecasting accuracy and adapting to complex environments [12]. Consequently, they have emerged as a major development trend in WPF in recent years. Accordingly, this review primarily focuses on DL methods within the broader machine learning paradigm. Fig. 4 illustrates a representative deep representation learning-based framework for WPF, highlighting its key components and workflow.

Recent studies have provided comprehensive reviews of DL and optimization methods in WPF. Wang et al. [2] compared the strengths, limitations, and computational costs of various deep neural networks (DNNs), and reviewed the applications of optimization algorithms in configuring DNNs, outlining future directions such as multi-objective optimization and cross-scale applications. Lu et al. [13] examined metaheuristic algorithms, highlighting the role of optimization layers in improving forecasting performance and systematically comparing different approaches in terms of accuracy, stability, and computational efficiency. Wu et al. [14] offered an in-depth overview of recurrent neural networks (RNNs), convolutional neural networks (CNNs), and numerous long short-term memory (LSTM)-based variants and hybrid models, emphasizing the advantages of DL in feature extraction and modeling complex nonlinear relationships. From a methodological perspective, Xie et al. [15] distinguished between deterministic and probabilistic forecasting paradigms, underscoring the importance of probabilistic forecasting for grid scheduling and risk management, and calling for greater integration of uncertainty quantification with DL methods. Liu et al. [16] reviewed the development and emerging trends of DL in WPF, stressing the potential of novel architectures such as generative adversarial networks (GANs) and Transformers, as well as cross-domain techniques like transfer learning and reinforcement learning, while emphasizing the need to advance interpretability and real-world applicability. Yang et al. [17] synthesized the scientific problems underlying different forecasting methods, highlighted the logical connections among them, and stressed the importance of data engineering, reproducibility, and interpretability in driving the field forward. A comparative summary of the scope and focus of these representative reviews is provided in Table 1.

The aforementioned reviews summarized the research progress in the field of WPF from different perspectives. However, with the rapid development of artificial intelligence, forecasting methods in this field are continuously evolving, and a wide range of emerging models have been introduced into related studies. In particular, methods based on graph neural networks (GNNs) and pre-trained large models have attracted growing attention and demonstrated superior performance. Therefore, it is necessary to systematically review and synthesize these recently emerging representative approaches.

This paper provides a comprehensive review and comparison of high-quality studies in this field over the past five years, with the aim of offering insights and references for future research and applications. To ensure focused and up-to-date coverage of deep representation-based WPF studies, this review adopted a structured literature search and screening strategy. Google Scholar was used as the primary search platform because of its broad coverage, while Web of Science was consulted as a supplementary source to cross-check the representativeness and quality of the retrieved literature. The literature search was primarily centered on the topic of “wind power forecasting”, and several related expressions, such as “wind power prediction”, “deep learning for wind power forecasting”, and “neural network wind power forecasting”, were additionally consulted to reduce the risk of missing representative studies. The search was conducted on a year-by-year basis from 2020 to the most recent available period, in order to capture recent methodological developments in deep representation learning-based WPF. In addition, a limited number of representative studies published before 2020 were also considered when they were deemed seminal or necessary for methodological context and comparison. Given the large volume

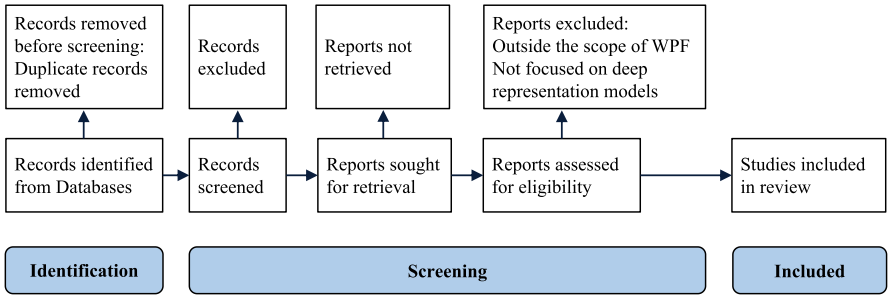


Fig. 2. Flowchart of the literature selection process.

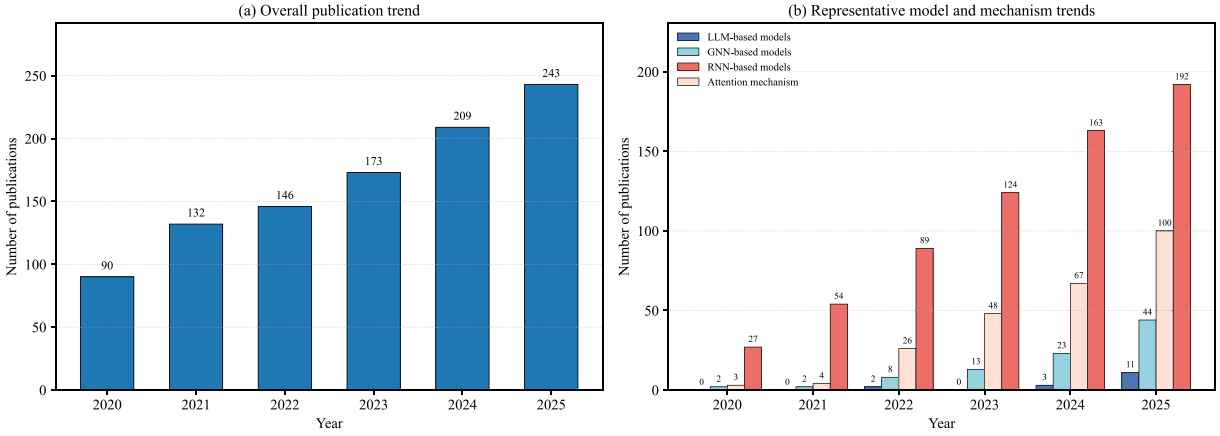


Fig. 3. Publication trends of wind power forecasting studies from 2020 to 2025: (a) overall publication trends; (b) representative model and mechanism trends.

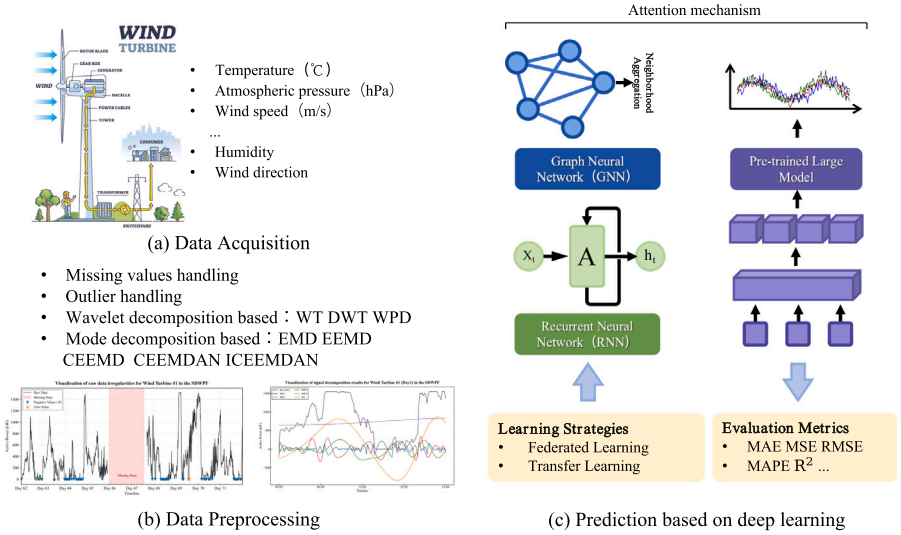


Fig. 4. Wind power forecasting framework based on deep learning.

of records returned by Google Scholar, the screening process was designed to focus on the most relevant and representative results in each yearly interval. Overall, the current bibliography contains more than one hundred entries, including several reviews and two reports. The retrieved literature was subsequently screened based on topic relevance, methodological scope, journal quality, and completeness of technical description. To ensure the quality and representativeness of the reviewed studies, priority was given to papers published in leading journals in the field. The final set of studies was then categorized according to methodological characteristics, including model architecture, data preprocessing strategy, hybrid design, and forecasting setting, forming the basis for the comparative analysis in this review. Inspired by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement [18], we provide a methodology flowchart for the paper selection process, as shown in Fig. 2, to summarize the identification, screening, eligibility assessment, and final inclusion of studies.

To further illustrate the recent research trend in WPF, we conducted a year-by-year publication statistics analysis based on Web of Science topic searches using relevant WPF-related keywords. As shown in Fig. 3,

Table 1
Summary of representative reviews.

Review	Year	Main focus	Data preprocessing	Prediction methods	Experimental analysis
Wang et al. [2]	2021	DNN-based WS/WP forecasting methods	Signal processing methods; Outlier detection methods	AE-based; RBM-based; CNN-based	Numerical case studies for different methods
Lu et al. [13]	2021	Meta-heuristic optimization for WPF	Data decomposition	Artificial neural networks; Classical machine learning	Comparison of performance of meta-heuristic algorithms
Wu et al. [14]	2022	DL architectures for WS/WP forecasting	/	RNN-based; RBM-based; CNN-based; AE-based models	/
Xie et al. [15]	2023	Deterministic vs probabilistic WS/WP forecasting	Data decomposition; Dimensionality deduction; Data denoising	Deterministic forecasting methods; Probabilistic forecasting methods	Evaluation metrics for deterministic and probabilistic models
Liu et al. [16]	2024	DL-WPF developments and trends; Problem formulation	Signal processing component; Clustering component	Advanced DL methods: 28 methods divided into 8 types	Optimization component; Feature engineering component
Yang et al. [17]	2025	Mathematical formulations; Scientific problems; Logical relations	Data decomposition; Data denoising	GNNs; Attention mechanism; Residual networks; Transfer learning; Incremental learning; Federated learning	/
Ours	2026	Advanced deep representation learning for WP forecasting	Data cleaning; Data denoising; Signal decomposition methods	LLM-based; GNN-based; RNN-based; Attention mechanism; Transfer learning; Federated learning	Details of representative experimental methods; Application-oriented analysis; Historical and prediction horizons

Abbreviations: WS: wind speed; WP: wind power; AE: autoencoder; RBM: restricted Boltzmann machine; CNN: convolutional neural network; RNN: recurrent neural network; GNN: graph neural network; LLM: large language model.

the number of WPF-related publications has increased steadily from 2020 to 2025, indicating sustained and growing research attention in this field. In addition to the overall growth trend, representative deep representation models and mechanisms also exhibit different development patterns. RNN-based models remain the most widely studied category, while GNN-based models show a continuous increase due to their ability to model spatial dependencies among turbines and wind farms. Attention mechanisms have also been increasingly incorporated into WPF models, reflecting their importance in temporal, feature, and spatial representation learning. Although methods based on large language models (LLMs) are still at an early stage, their emergence in recent years suggests a new direction for deep representation-based WPF.

The main contributions of this work are as follows:

- This review systematically surveys representative studies in recent years that are based on pre-trained large models, GNNs, and RNNs. In particular, it provides a structured synthesis of approaches involving pre-trained large models, which, to the best of our knowledge, is the first systematic summary of this line of research in existing WPF reviews. In addition, it offers a comprehensive overview of the roles of attention mechanisms across different dimensions.
- Data preprocessing methods in WPF are systematically and comprehensively reviewed, with particular attention to signal decomposition-based methods. This review not only summarizes the fundamental principles, strengths, and limitations of these methods, but also traces their developmental trajectory, thereby offering a structured reference framework for subsequent model construction and methodological optimization.
- The role of transfer learning in cross-scenario knowledge transfer, as well as the advantages of federated learning in privacy preservation and multi-wind-farm collaborative forecasting, are summarized. In addition, the available public datasets are reviewed, the experimental details of representative methods were summarized, and application scenarios were analyzed.

The remaining sections are organized as follows: Section 2 introduces typical issues and solutions in data preprocessing, and compares the strengths and weaknesses of various signal decomposition methods. Section 3 provides a systematic overview of deep representation-based forecasting models, covering representative approaches such as LLMs, GNNs, and RNNs, and highlights the roles of attention mechanisms across different dimensions. Section 4 summarizes the applications of transfer learning and federated learning in WPF, with emphasis on their potential for cross-scenario knowledge transfer and privacy-preserving collaborative modeling. Section 5 organizes and compares the experimental settings of representative methods, including available datasets, evaluation metrics, historical and prediction horizons, and application-oriented analyses under different scenarios. Section 6 discusses key research challenges and gaps from the perspectives of data, model development, and practical applications, and further outlines several emerging techniques.

2. Data preprocessing

Deep representation–based methods for WPF usually require a combination of historical data, generally including environmental factors (e.g., wind speed, wind direction, and air pressure) and turbine status (e.g., generator speed, blade pitch angle, and gearbox bearing temperature), which are usually captured and logged by supervisory control and data acquisition (SCADA) systems, which are widely equipped in modern wind turbines for remote monitoring and control [19,20]. Data preprocessing is a crucial step in WPF and is an important prerequisite to ensure model performance and accuracy. The primary processing operations comprise data cleaning, redundant information processing, and signal decomposition.

2.1. Data cleaning

In order to reduce the burden of data collection and transmission, SCADA systems usually collect various data in 10-minute or 15-minute intervals to meet the data requirements of WPF [21]. However, the data

collected by the system may be abnormal due to wind turbine maintenance shutdowns, sensor failure/degradation, or system errors [22]. In order to mitigate the impact of abnormal data on forecasting accuracy and improve computational efficiency, methods should be used to process abnormal data. Anomalous data mainly include two categories: missing values and outliers [23].

2.1.1. Missing value handling

The missing value problem is common in time series data. Generally, missing values represent a relatively small percentage of the overall data and can usually be filled in using simple methods. Gong et al. [24] used the previous time step data to fill in the missing values. Wang et al. [25] suggested that the data could be filled in using the B-spline interpolation algorithm without affecting the regularity or trend of the data. For zero values, Ye et al. [26,27] used the average of the two surrounding data points as a replacement. For continuous zero sequences, the data were replaced by linear interpolation. Qiu et al. [28] used a K-nearest neighbor imputation technique to fill in these missing values. Zhao et al. [23] filled the missing values by fitting the relationship between wind speed and active power using a polynomial. Tian et al. [29] used a tracking-removed autoencoder (TRAE)-based imputation scheme. Specifically, missing values are treated as variables in the TRAE loss function and are included in the network training process along with incomplete datasets to maximize the use of the available data, which are updated using an optimization algorithm. This approach allows for a more comprehensive use of the data rather than relying solely on known values. As iterative optimization proceeds, the estimation error of missing values is gradually reduced, resulting in improved model accuracy and imputation precision.

2.1.2. Outlier handling

Outliers refer to data points that deviate from the normal range. It is usually necessary to detect anomalies in sequence data, remove outliers, and then use appropriate methods to fill the corresponding positions to obtain clean sequence data. There are many ways to deal with outliers. Zhao et al. [23] used data fitting for outliers. Wang et al. [25] used statistical methods to detect and replace outliers. Zhao et al. [30] used robust regression to deal with outliers. Xing et al. [31] used kd-tree-accelerated DBSCAN as an anomaly detection method. Wang et al. [32] developed a new log-likelihood function as a loss function used in the training phase to alleviate the effect of outliers in the original data on forecasting accuracy and improve computational efficiency. Wang et al. [33] used isolation forest (IF) to detect outliers in a data sequence, the IF algorithm constructs a predefined number of iTrees and utilizes a binary search tree structure called isolation trees for isolating the samples, ultimately in order to form a completely isolated forest of anomaly-detecting data. Kisvari et al. [34] utilized the IF algorithm for detecting and deleting a variety of outliers. Specifically, IF is an ensemble learning method based on isolation trees and detects anomalies by recursively partitioning the data. By correcting abnormal data, the prediction accuracy of the model can be effectively improved.

2.2. Feature reduction and normalization

The data collected by SCADA systems are usually multi-scale data. However, multi-scale variables not only increase the difficulty of model training but may also introduce redundant information, leading to model overfitting. Therefore, it is necessary to process the redundant data to improve the training efficiency and accuracy of the model. Xiao et al. [35] used the Pearson correlation coefficient (PCC) to determine the influence of different input features on the actual output power, assigned weights to the input variables in each dimension, and used principal component analysis (PCA) for dimensionality reduction to improve model performance. Wang et al. [33] utilized manifold learning to remove data redundancy through dimensionality reduction to represent and process data more efficiently. Compared with the unified dimensionality reduction methods such as PCA, manifold learning can retain the nonlinear structure of the data in the process of dimensionality reduction, thereby better revealing the intrinsic properties of the data. Tian et al. [29] used the RReliefF algorithm to eliminate redundant data features and effectively filtered out features with weights below a predefined threshold through weight assignment. In addition, the maximal information coefficient (MIC) [36] is also a powerful tool for dealing with the relationship between variables. Given a two-variable dataset (T_i, T_j) with $D = \{(x_1, y_1), \ldots, (x_n, y_n)\}$, a scatter plot is first generated and then partitioned into grids in the X-Y coordinate space. $\mathrm{MIC}(T_i, T_j)$ can be calculated based on mutual information (MI). The detailed calculation process is omitted here for brevity. Sun et al. [21] utilized MIC for the assessment of spatial information to consider irrelevant wind data as noise in spatial information and to improve the reliability of knowledge about wind-condition relationships between turbines. More recently, Xu et al. [37] further combined grey relational analysis (GRA), PCC, and MIC to construct a comprehensive correlation indicator for feature selection, thereby reducing redundant inputs and retaining the most relevant variables for ultra-short-term WPF. This design also reflects the limitation of using a single correlation metric: GRA is more suitable for characterizing geometric similarity, PCC is effective for identifying linear relationships, and MIC is more capable of capturing nonlinear dependencies. By integrating these complementary indicators with equal weights, the method helps reduce the bias introduced by any single metric and enables a more balanced evaluation of feature relevance.

In addition, normalization techniques are widely used to reduce the differences in data scales and facilitate model convergence. Among them, min–max normalization is one of the most commonly adopted methods in previous studies, and its mathematical expression can be written as follows:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{1}$$

where x denotes the original value of a feature, x' denotes the normalized value, and $x_{\min}$ and $x_{\max}$ denote the minimum and maximum values of the corresponding feature, respectively. Through this transformation, all input features, including wind speed and wind direction, are scaled to the interval $[0, 1]$, which helps balance differences in variable scales and improve training efficiency [26,38].

2.3. Signal decomposition methods

Due to the uncertainty of environmental conditions around wind turbines, the sequence data collected by SCADA systems typically exhibit randomness and fluctuation, resulting in highly nonstationary and nonlinear characteristics. Therefore, many studies employ signal decomposition techniques to break down the raw sequence into several smoother sub-sequences before forecasting. By doing so, the complex prediction task is transformed into modeling simpler, more predictable stationary series. To facilitate understanding, Fig. 6 illustrates the decomposition results of actual operational data using two representative methods: empirical mode decomposition (EMD) and discrete wavelet transform (DWT). Both methods effectively isolate the chaotic high-frequency components from the underlying smooth operational trends. This isolation process significantly reduces the representation learning difficulty for subsequent deep prediction models.

2.3.1. Wavelet-based methods

Traditional signal theory is based on Fourier analysis. However, for nonstationary signals, the Fourier transform cannot effectively reflect the change in its frequency over time. Wavelet transform (WT) [40], as a local time-frequency transformation, decomposes the original time series into low-frequency and high-frequency components, and further decomposes the low-frequency components into another level of low-frequency components (Fig. 5(c)). In this way, useful information can be effectively extracted from the signal, making WT widely used in

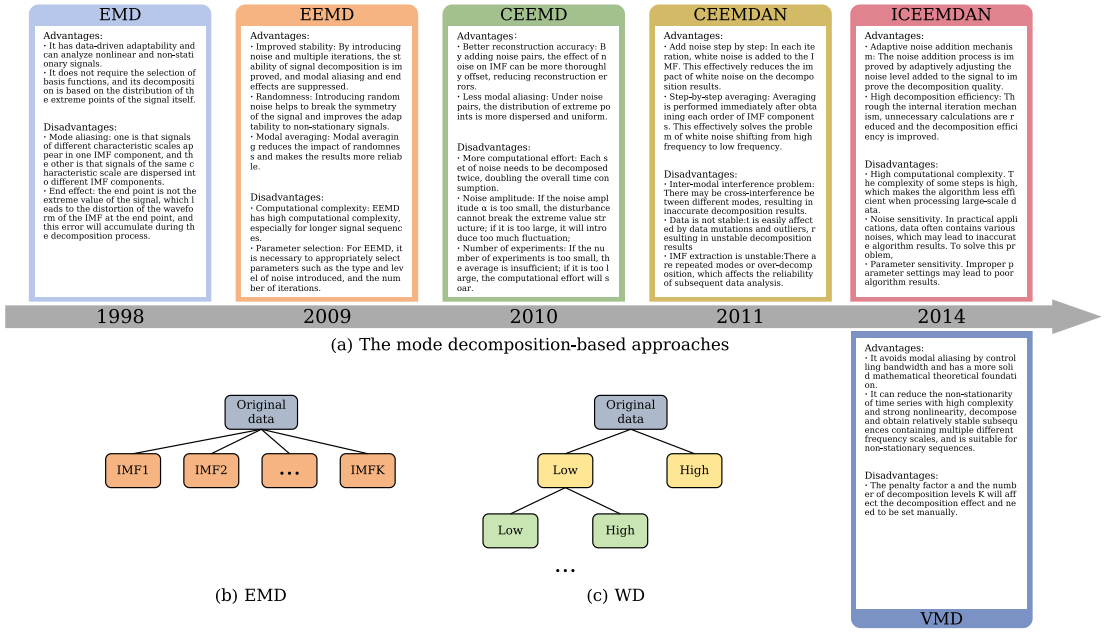


Fig. 5. The evolution and structural comparison of data decomposition techniques for WPF.

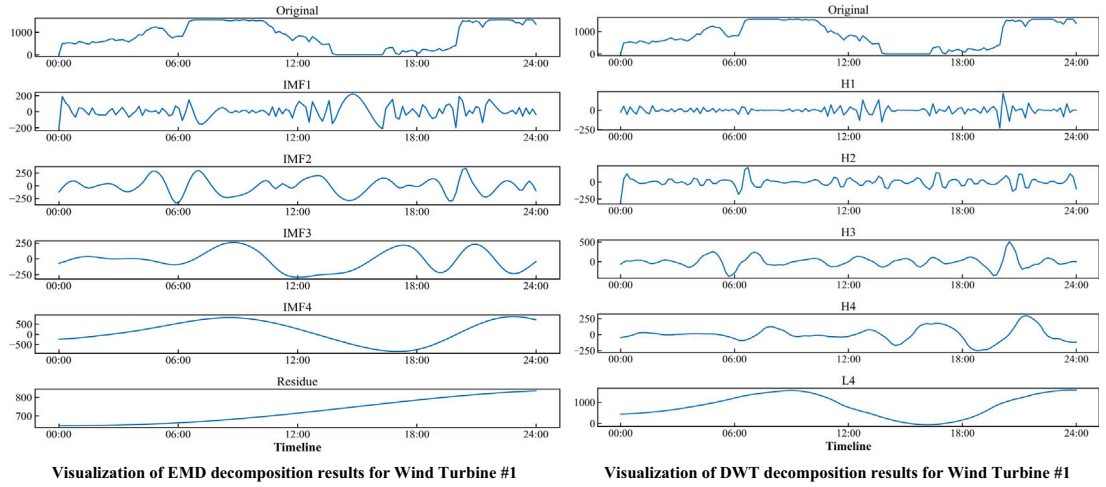


Fig. 6. Signal decomposition results of wind turbine #1 operational data: (Left) EMD and (Right) DWT. The original data comes from the SDWPF dataset [39].

sequence prediction tasks such as load forecasting and WPF. DWT performs signal analysis by discretizing the scale and translation parameters of the mother wavelet, which significantly reduces the computational complexity and the redundancy of the time-frequency analysis results by limiting the discrete scale and displacement parameters compared to the continuous wavelet transform (CWT). Since many wavelet families can be used in DWT, finding the optimal mother wavelet is crucial for machine learning methods to achieve the best performance. In order to determine which type of wavelet shows higher relevance in meteorological signal reconstruction, Nascimento et al. [41] compared the approximate first-order reconstructed signal after wavelet decomposition with the original signal and performed statistical evaluation through root mean square error (RMSE). Traditional DWT ignores the decomposition in the high-frequency band, which may cause the loss of important signal details in the application of WPF. Wavelet packet decomposition (WPD) is based on WT. In addition to further decomposing the low-frequency sub-band, it also further decomposes the high-frequency sub-band at each level of signal decomposition [42]. Based on this, the high-frequency information of the signal can be fully utilized. The performance of WPD depends largely on the selected mother wavelet and decomposition level. Hanifi et al. [43] selected 16 wavelets from four widely used wavelet families (Daubechies, Haar, Sym, and Coif) and evaluated their performance in improving the accuracy of prediction models, including linear regression, random forest, feed-forward neural network, and LSTM. The results showed that the fifth-order Daubechies wavelet had the best performance and was therefore selected as the mother wavelet in this study.

2.3.2. *Mode decomposition-based methods*

The methods based on EMD are widely used in WPF because of their obvious advantages in dealing with nonstationary and nonlinear data. Fig. 5(a) systematically illustrates the developmental trajectory, advantages, and limitations of these mode decomposition-based approaches. Unlike wavelet analysis, the most notable feature of EMD is that it overcomes the non-adaptability problem of basis functions. As shown in Fig. 5(b), for an unknown signal, EMD can directly decompose it into the

sum of several intrinsic mode functions (IMFs) without prior analysis, which is mathematically expressed as: $x(t) = \sum \mathrm{IMF}_i(t) + r(t)$. Yuzgec et al. [44] used EMD as an important component of the comprehensive WPF model, decomposed wind speed and power time series into IMFs, captured high-frequency and low-frequency information in a data-driven manner, reduced the noise of the original data series to a certain extent, and improved the effectiveness of WPF. However, the EMD method suffers from the problem of mode aliasing. To address this issue, Huang et al. [45] proposed ensemble empirical mode decomposition (EEMD), a noise-assisted signal analysis method. Specifically, white noise of finite amplitude is added to the original signal to modify the extreme points in the signal, and IMF takes the average value of multiple decomposition results. This modal averaging helps to reduce the impact of randomness, improve the stability of the results, and maintain the physical uniqueness of the decomposition [38]. However, for longer signal sequences, the computational complexity of EEMD is higher, and parameters such as the type and level of introduced noise and the number of iterations need to be chosen appropriately. Yin et al. [27] used the complete ensemble empirical mode decomposition (CEEMD) method to add a pair of opposite positive and negative white noises as auxiliary noises into the source signals, in order to eliminate the redundant auxiliary white noises in the reconstructed signals after decomposition by the original EEMD method, and at the same time to reduce the number of iterations required for decomposition, thus lowering the computational cost. Although EEMD and CEEMD solve the modal aliasing problem to a certain extent, they may still leave a certain amount of white noise in the decomposition results. In addition, the introduction of white noise may lead to an inconsistent number of modes in the signal, thus affecting the subsequent signal analysis. Hossain et al. [46,47] used the complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) algorithm. By introducing step-by-step noise and step-by-step averaging processing, CEEMDAN can effectively reduce the impact of noise on signal decomposition and show better signal decomposition and noise reduction capabilities. Moreover, CEEMDAN has also been incorporated into recent advanced forecasting frameworks. For example, in a multi-scale prompting framework for WPF with LLMs, CEEMDAN was used to decompose wind power series into multi-scale components before downstream modeling, further demonstrating its applicability beyond conventional forecasting architectures [48]. Improved complete ensemble empirical mode decomposition with adaptive noise (ICEEMDAN) [49], as a modification of CEEMDAN, is able to extract vibration modes in signals more accurately by selecting the IMF component of noise to control the degree of noise introduction. In contrast, ICEEMDAN has more accurate decomposition results, noise control, and applicability [46,50].

In signal processing, variational mode decomposition (VMD) [51] is also an effective signal decomposition method. The core idea of VMD is to formulate and solve a constrained variational problem for adaptive signal decomposition. The corresponding variational formulation can be expressed as

$$\min_{\{u_k\},\{\omega_k\}} \left\{ \sum_{k=1}^{K} \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] \exp(-j\omega_k t) \right\|_2^2 \right\} \quad \text{s.t.} \quad \sum_{k=1}^{K} u_k(t) = f(t) \tag{2}$$

where $f(t)$ denotes the original input signal, $u_k(t)$ denotes the k-th decomposed mode, ω_k denotes the center frequency of the k-th mode, K is the total number of modes, t denotes time, $\delta(t)$ is the Dirac delta function, j is the imaginary unit, $*$ denotes the convolution operator, ∂_t denotes the partial derivative with respect to time, and $\|\cdot\|_2^2$ denotes the squared L_2-norm. Compared with EMD, VMD has a more rigorous mathematical foundation. It redefines the IMF as a band-limited intrinsic mode function under stricter constraints, i.e., $u_k(t) = A_k(t)\cos(\phi_k(t))$. The optimal solution of the variational model is obtained iteratively, through which the center frequency and bandwidth of each mode are determined. Therefore, VMD is a fully non-recursive decomposition method. Although VMD alleviates the modal aliasing problem of EMD by controlling the bandwidth, its performance is affected by the preset modal number K. In order to determine the optimal modal number K, Zhao et al. [30,52] analyzed the correlation of adjacent modal components at different modal numbers, and searched for the optimal parameter based on whether the center frequency was similar at different numbers of IMFs. Once a similar frequency appeared from the k-th IMF, k-1 was chosen as the optimal decomposition number. Rayi et al. [53] used a newly presented meta-heuristic population based sine cosine integrated water cycle algorithm to determine the optimal value of parameter K. In addition, VMD has also been integrated with optimized ensemble learning frameworks for WPF. For example, VMD and bagging extreme learning machine with multi-objective optimization for WPF [54] combined VMD with a bagging extreme learning machine and a multi-objective optimization strategy, showing that decomposition-based preprocessing can be further enhanced through coordinated model design and parameter optimization. Moreover, VMD has also been integrated with other preprocessing and uncertainty-aware forecasting techniques. For example, a recent hybrid framework combined VMD with singular spectrum analysis (SSA) for data preprocessing, and further incorporated conformal prediction and SHAP-based feature selection, indicating that VMD can also support interpretable and uncertainty-aware wind forecasting frameworks [55]. Multivariate VMD (Multi-VMD) [56], as a multivariate extension of the original VMD algorithm, can be used to process multivariate data containing any number of data channels and align the frequency information across multiple data channels in each IMF. Yang et al. [57] used Multi-VMD to realize the time-frequency synchronization analysis of wind power series and multidimensional meteorological series, which solved the limitation of general signal decomposition strategies that only decompose the wind power series itself and improved the robustness of the prediction model. Beyond standalone signal decomposition, VMD can also be embedded into hybrid forecasting frameworks. Romero-Barrera et al. [58] integrated VMD with analytical wake models and several DL architectures, where wake-model outputs and historical forecast deviations were jointly used as input features. Their results showed that VMD preprocessing further improved forecasting accuracy, particularly for short- and medium-term horizons.

3. Deep representation prediction models

To provide a clearer overview of representative deep representation-based models for WPF, this section discusses three main model architectures, namely LLM-based models, GNN-based models, and RNN-based models, as shown in Fig. 7. In addition, since attention mechanisms have been widely incorporated into various forecasting frameworks, this section also discusses their roles across different dimensions.

3.1. Attention mechanism

With the widespread adoption of DL in WPF, attention mechanisms have emerged as a key technique for improving forecasting accuracy and model interpretability, due to their advantages in modeling complex temporal data. Researchers have introduced attention mechanisms into different dimensions according to problem characteristics, which can be broadly categorized into four types: temporal, feature, spatial, and others (summarized in Table 2).

From the temporal perspective, attention mechanisms are used to capture the differentiated contributions of individual historical time steps to forecasting outcomes. Ye et al. [59] proposed the Seq2Seq-Attention model, which dynamically assigns temporal weights to effectively enhance the modeling of sequence dependencies. Tian et al. [46] incorporated dual-stage self-attention into the ST-LSTM framework, assigning weights at both the input and hidden layers to improve the capture of long-range dependencies. Dai et al. [60] proposed an

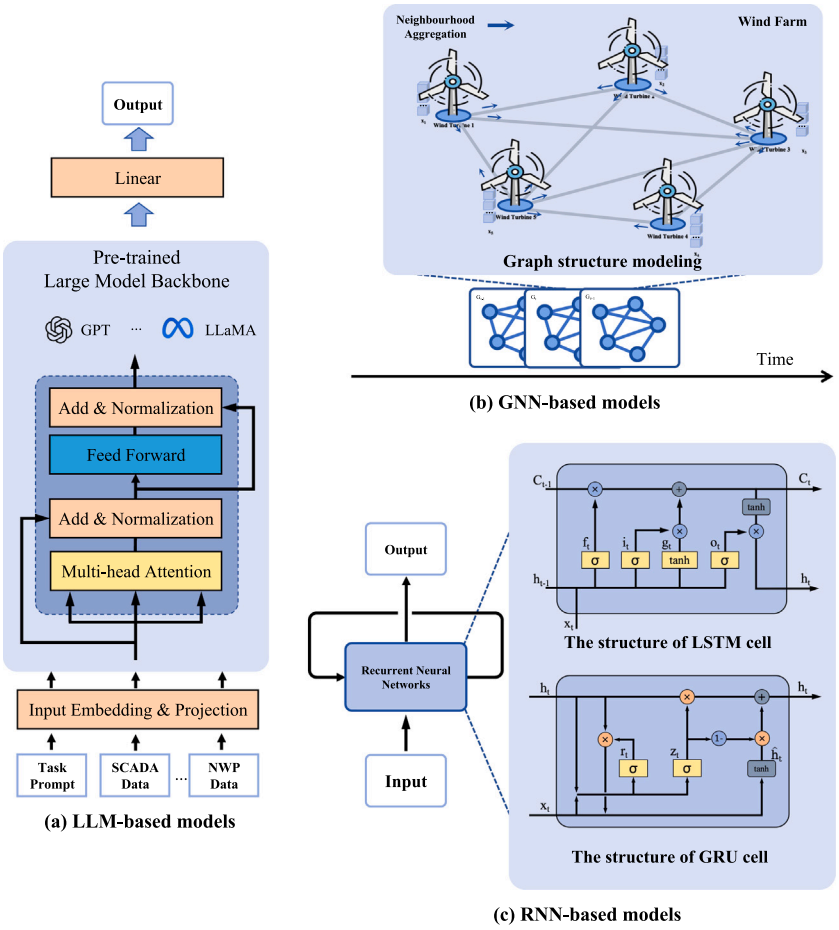


Fig. 7. Overview of representative deep learning architectures for WPF: (a) LLM-based models, (b) GNN-based models, and (c) RNN-based models.

online self-attention approach that adaptively adjusts the historical horizon, thereby improving both the accuracy and adaptability of ultra-short-term forecasting. Arooj et al. [61] developed FedWindT, which leverages the self-attention mechanism within a Transformer architecture to enable cross-wind-farm collaborative forecasting. Wang et al. [62] introduced the Temporal Fusion Transformer, which integrates static features with historical power sequences and achieves higher accuracy in ultra-short-term forecasting. In addition, Liu et al. [63] proposed the Bayesian BiGRU-FT-Attention model, which combines a feature-attention module and a temporal-attention module (FT-Attention) to highlight critical time segments, thus further improving forecast stability and interpretability.

WPF often involves multi-source inputs such as wind speed, wind direction, meteorological parameters, and turbine status, with significant differences in the relative importance of these features. Against this background, incorporating attention mechanisms can effectively identify and highlight key features, thereby enhancing the representational capacity of forecasting models. Zhang et al. [64] proposed the Multi-Source and Temporal Attention Network, which employs a variable attention module to identify key features in numerical weather prediction (NWP) data and incorporates temporal attention to model historical sequences, thereby improving probabilistic forecasting performance. Hu et al. [65] developed an attention-enhanced LLM framework that jointly models target and non-target variables through multiple attention mechanisms, thus overcoming the limitations of manual feature selection. Similarly, Sun et al. [66] incorporated attention mechanisms into a multi-factor extraction framework to dynamically weight input features, enabling more accurate feature selection and integration. In addition, Liu et al. [63] proposed the Bayesian BiGRU-FT-Attention model, which also operates at the feature dimension. By focusing on the most relevant meteorological and power variables and integrating attention with Bayesian uncertainty modeling, this approach not only improves prediction accuracy but also quantifies the reliability of forecasting results. These approaches demonstrate that, at the feature dimension, attention mechanisms facilitate multimodal data fusion and the extraction of key driving factors.

Significant spatial dependencies exist among turbines within a wind farm and across different wind farms. Attention mechanisms can be employed to model these spatial correlations and effectively characterize inter-node relationships. Lai et al. [67] proposed the BERT4ST model, which applies bidirectional attention to capture the spatiotemporal coupling of wind farms. Yang et al. [57] combined CNN-BiLSTM with multi-head attention (MHA) to capture global features and long-range dependencies of regional wind fields. Hou et al. [68] incorporated a graph attention mechanism into a multi-graph neural network and Transformer framework to jointly model multiple spatial relationships (adjacency, functional similarity, and geographic topology), thereby achieving high-accuracy multi-step spatiotemporal forecasting. Song et al. [69] embedded self-attention into a multi-resolution convolutional network to enhance both local and global spatial feature modeling. In addition, Liu et al. [70] combined CNN, GNN, and self-attention to adaptively model inter-turbine spatial relationships, demonstrating stronger robustness under noisy and missing data conditions. Moreover, graph attention networks (GATs) [23] have been widely adopted to dynamically adjust the weights among turbine nodes, thereby capturing both global and local dependencies in cluster forecasting. These approaches

Table 2
Different roles of attention mechanisms.

Dimensions	Ref.	Role
Temporal dimension	[46,59–63]	a) Distinguishes the contributions of different historical moments to future power generation and highlights critical time segments; b) Effectively captures long-range dependencies, mitigating gradient vanishing and information loss problems in traditional RNNs/LSTMs when handling long sequences; c) Improves the accuracy and stability of ultra-short-term and short-term forecasting.
Feature dimension	[63–66]	a) Dynamically assigns weights among multi-source inputs; b) Identifies the most valuable features for forecasting while suppressing redundant and noisy information; c) Enhances feature fusion efficiency and improves model interpretability.
Spatial dimension	[23,57,67–70]	a) Captures spatial dependencies among turbines within and across wind farms; b) Dynamically adjusts the strength of connections between nodes or regions, capturing both local and global spatial features; c) Improves the accuracy and robustness of large-scale wind farm cluster forecasting.
Others	[50,71–74]	a) Lag correction: Aligns the time delay between NWP data and actual observations, reducing prediction bias caused by asynchrony; b) Uncertainty modeling: Combines attention with Bayesian or probabilistic methods to quantify the reliability of forecasting results; c) Cross-modal fusion: Serves as a feature-weighting mechanism when integrating wind power data with text, images, or other modalities, enhancing cross-modal forecasting and interpretability; d) Generalization enhancement: Integrates adversarial training or transfer learning to improve model adaptability under distribution shifts.

highlight the unique value of attention mechanisms in the spatial dimension, offering new technical pathways for cross-turbine and cross-farm forecasting.

In addition to weighted modeling across temporal, spatial, and feature dimensions, attention mechanisms have also been extended to tasks such as lag correction, cross-modal learning, and uncertainty modeling. Niu et al. [50] proposed Dual-Stage Attention (DSA), which assigns weights at both the input and hidden layers, thereby enhancing forecasting robustness and interpretability. Their subsequent work [71] integrated trend–seasonality representation learning with a Temporal Fusion Transformer, using attention weights to reveal the relative importance of different input features. Zhu et al. [72] developed the CMLLM model, which applies cross-modal transfer to convert wind power data into textual representations and incorporates an attention prompting module to activate the reasoning capability of LLMs. Addressing the time-varying characteristics of wind power sequences (i.e., varying power correlations across months or time periods), Liu et al. [73] proposed Temporal Lag Attention, which effectively corrects the lag in NWP. Tang et al. [74] employed a transfer learning approach that integrates attention mechanisms with adversarial training to improve generalization under distribution shifts.

Overall, the application of attention mechanisms in WPF exhibits multi-dimensional and diversified development trends. Across temporal, feature, and spatial dimensions, attention mechanisms substantially improve forecasting performance; in extended applications such as lag correction, cross-modal fusion, and uncertainty modeling, their potential is also becoming increasingly evident. Future research could further explore cross-dimensional attention integration and enhancements in interpretability, thereby advancing WPF in terms of accuracy, robustness, and practical applicability.

3.2. LLM-based models

In recent years, LLMs have emerged as a major development direction in the field of artificial intelligence. With their massive parameter sizes and powerful representational capacity, these models have demonstrated outstanding performance across a wide range of tasks [75,76]. Such models typically adopt deep neural network architectures and are pre-trained on massive datasets, enabling them to learn rich semantic representations and capture complex dependencies in long sequential data [77]. As the technology continues to mature, research on applying LLMs to time series tasks is experiencing rapid growth.

Lai et al. [67] proposed BERT4ST, a BERT-based spatiotemporal prediction model, which, to our knowledge, is the first application of a fine-tuned pre-trained large language model (LLM) in WPF. They used a pre-trained BERT encoder as a backbone network to learn individual spatiotemporal dependencies of wind power data patches. Experimental results show that BERT4ST achieves ideal performance compared to some state-of-the-art methods. Following this line of research, Yu et al. [78] proposed the Wind2vec-BERT model to further adapt the Transformer architecture for continuous wind data. Unlike direct tokenization, they introduced a Wind2vec embedding mechanism—inspired by Time2vec—to map wind speed and direction into continuous vector representations with periodic activation functions, thereby better capturing the inherent temporal dynamics. To address the computational redundancy of large models, they integrated the Adaptive Computation Time (ACT) algorithm into the BERT backbone, allowing the model to dynamically adjust calculation steps based on data complexity. Furthermore, a GARCH-M model was employed for post-processing to rectify prediction residuals, significantly enhancing robustness against high-volatility wind conditions. LLMs are inherently adept at processing text, but have limited ability to directly interpret numerical time series data. To address this limitation, Fan et al. [79] proposed a semantic transformation mechanism for time series data, introducing two key embedding modules: the Prompt Embedder and the Data Embedder. The Prompt Embedder converts task descriptions, historical data features, and numerical weather prediction (NWP) information into embedding vectors interpretable by LLMs, while the Data Embedder transforms numerical time series into an LLM-compatible format. This design ensures that LLMs can simultaneously leverage the “task guidance” provided by textual prompts and the “historical patterns” embedded in numerical data, thereby enhancing forecasting accuracy. To adapt pre-trained LLMs to the specific task of WPF while avoiding the high cost of full-parameter fine-tuning, the study employed a parameter-efficient fine-tuning strategy known as Low-Rank Adaptation (LoRA) to improve adaptation efficiency. Experimental results confirmed that M2WLLM achieved high predictive accuracy, strong few-shot learning capability, robustness, and semantic understanding ability. Duan et al. [80] designed a soft and hard prompt fusion mechanism to incorporate WPF-specific task information into LLMs, further stimulating the zero-shot learning ability of LLMs, thereby achieving flexible predictions in different prediction ranges. Building on this prompt-based paradigm, they subsequently proposed an Attentive Residual Prompt Tuning (ARPT) approach [81] to address the challenge of missing operational data. By reconstructing the hard prompt generator to accommodate incomplete inputs and integrating a residual self-attention adapter with bridge-shared parameters, this method significantly enhances the robustness and extrapolation capability of LLMs under highly volatile and sparse data conditions. Based on the limitations of conventional LLMs in time series forecasting, namely their reliance on prompt engineering and insufficient capability in capturing cross-channel information, Hu et al. [65] proposed a framework named MultiAttLLM. This model employs a pre-trained LLM as its core

component and incorporates cross-attention and self-attention mechanisms to effectively model multivariate time series data. In the model selection process, several candidates were compared, including BERT, GPT-2, LLaMA-3.2-1B, and LLaMA-3.2-3B. GPT-2 was ultimately chosen because it offers the best balance between prediction accuracy, memory usage, and training efficiency, making it suitable for a single GPU environment. Furthermore, to bridge the gap between textual data processing and multivariate time series forecasting, the method uses a word projection module and cross-attention to transform time series into a format interpretable by the LLM. This design eliminates the need for manual prompt construction in conventional LLM applications. Experimental results show that MultiAttLLM delivers high predictive accuracy and strong generalization capability across diverse regions and climatic conditions. Zhu et al. [72] first applied a cross-modal LLM-based method to WPF, using data cross-modality and pre-trained LLMs, which are highly compatible with BERT, GPT, and LLaMA, and can adapt to data with different characteristics. By adopting a cross-modal transfer learning method to fully process the data, the data is converted to the text modality, eliminating the need to retrain or fine-tune LLMs, while effectively alleviating the decline in prediction accuracy due to modality mismatch. While Zhu et al. focused on avoiding parameter updates, Zheng et al. [82] explored the potential of Parameter-Efficient Fine-Tuning (PEFT) to unlock the value of LLMs in downstream decision-making. They employed LoRA to fine-tune the LLaMA model for WPF via supervised instruction tuning. Beyond improving prediction accuracy, they innovatively constructed a risk-tunable uncertainty set based on the LLM's prediction errors (Wasserstein distance). This design integrates the LLM into a robust optimization framework for microgrid scheduling, effectively bridging the gap between high-precision forecasting and reliable energy management. Along this line, Meng et al. [83] proposed PowerMistral, a data-efficient WPF framework built on Mistral-7B for few-shot scenarios. Through a two-stage fine-tuning strategy, together with a gated multi-scale convolutional neural network (GMCNN), advanced temporal encoding, and trend-aware textual prompts, the model effectively captures both short-term fluctuations and long-term dependencies in wind power sequences. In addition, wavelet-based pattern decoupling was introduced to enhance the representation of multi-scale temporal structures, and experiments on six operational wind farms demonstrated strong performance under small-sample conditions. To further address the nonstationarity and multi-scale dynamics inherent in wind power data, Liu et al. [48] proposed the CPLLM-WPF framework. Unlike methods that process raw sequences directly, this approach integrates a CEEMDAN-based decomposition strategy to disentangle complex wind patterns into intrinsic mode functions (IMFs). By combining cross-modal reprogramming with task-aware prompts and employing LoRA for parameter-efficient fine-tuning, the model effectively captures both long-term trends and high-frequency fluctuations, demonstrating superior generalization capabilities in zero-shot scenarios across spatially heterogeneous wind farms.

Foundation large time series models (LTSMs) are specifically designed for time series data and are pre-trained on large-scale, multi-domain time series datasets, endowing them with enhanced time series analysis capabilities and cross-domain generalization. This makes them particularly suitable for time–series-intensive tasks such as WPF. Focusing on foundation LTSMs, Fan et al. [84] introduced the concept of Time Series Prompt (TSP) to enable flexible adaptation of foundation LTSMs to downstream tasks without modifying the backbone network. TSP comprises two forms: Exogenous Prompt (EX) and Prompt Tuning (PT). EX embeds exogenous time series (e.g., wind speed forecasts) into a high-dimensional vector space and introduces a Normalization Bias (NB) to preserve the physical magnitude of the target sequence. It then aligns patterns via a multi-layer perceptron to generate a prompt vector, which is concatenated with the target variable embedding sequence and fed into the model to guide forecasting. PT draws inspiration from the concept of soft prompts in LLMs, constructing trainable virtual tokens that are injected layer-by-layer into the Transformer's input. Combined with NB to preserve dimensional information, this approach enables parameter-efficient fine-tuning. The two methods can be applied independently or jointly. With the backbone kept frozen, only prompt-related parameters are trained (approximately 0.63M for EX and 8.96M for PT), thus achieving a balance between computational efficiency and forecasting performance.

3.3. GNN-based models

Graph neural networks (GNNs) are DL methods based on graph structures. In recent years, the use of GNNs for WPF has become increasingly popular, among which graph convolutional networks (GCNs), graph attention networks (GATs), spatiotemporal graph neural networks (STGNNs) and adaptive graph convolutional recurrent networks (AGCRN) are the mainstream research methods. Compared with traditional neural networks such as CNNs and RNNs, GNNs can effectively model the complex relationships between nodes with the help of the adjacency matrix of the graph, and integrate the information of adjacent nodes into the target node through the message passing mechanism to generate a new representation of the node. Its simplified mathematical description is as follows:

$$Z = \sigma(AXW) \tag{3}$$

where X is the node representation, W is the weight matrix, which is used to project the original node representation into the new space through linear combination, A is the adjacency matrix, σ represents the activation function, and Z is the new representation obtained by neighborhood aggregation.

The spatial correlation between the input variables plays a crucial role in WPF. However, some previous studies have neglected the spatial dependence of input variables, leading to unsatisfactory prediction results. Beyond correlation modeling, graph-based methods have also been used to capture causal relationships among meteorological variables and wind power outputs [85]. As mentioned above, the relationships among wind power and related variables can be represented by an undirected graph, whose adjacency matrix is defined as

$$A_{ij} = \begin{cases} w_{ij}, & \text{if } (v_i, v_j) \in E, \\ 0, & \text{otherwise}, \end{cases} \tag{4}$$

where A_{ij} denotes the element in the i-th row and j-th column of the adjacency matrix A, v_i and v_j denote the i-th and j-th nodes, respectively, E denotes the edge set of the graph, and w_{ij} denotes the weight of the edge between node v_i and node v_j, reflecting the strength of their relationship.

3.3.1. GCN

The GCN model [86] proposed in 2017 can be regarded as the pioneering work of GNNs and the basis of many complex GNNs. It was among the first methods to extend convolution operations from image processing to structured graph data.

Song et al. [69] proposed a novel MIC-based GCN prediction method. They not only determine the weights of different nodes using MIC, but also propose a new method to learn dynamic adjacency matrices in each training iteration. As training samples increase, the correlations between the input variables become more precise. After constructing the node representation and adjacency matrix, the GCN is used to integrate information from other neighboring nodes into the target wind power node. The GNN-based modeling approach in WPF is flexible enough to construct relationship graphs between input variables to mine their correlation features, as well as to model the historical data of different wind turbines in a single wind farm or different wind farms in a wind cluster. For cluster-level WPF, Xu et al. [87] proposed a prediction method based on feature selection and optimal graph structure GCN. This approach first performs feature selection on wind farm clusters to reduce the complexity and computational cost of the prediction model. Subsequently,

based on the results of feature selection, the number of nodes and features is optimized, and the optimal graph structure is constructed by considering dynamic updating of the adjacency matrix. Finally, the constructed optimal graph structure is used as an input to the GCN. Since the spatiotemporal interdependencies of WPF are closely related to the complex relationships between turbines, it is not enough to rely on a single or fixed method to construct the adjacency matrix. Hou et al. [68] proposed a multi-graph fusion GCN method, which constructed multiple graphs to represent the inter-turbine correlations driven by different factors, and then integrated the node information from each graph through an attention mechanism to obtain a unified graph, and then performed convolution in the spectral domain, namely spectral graph convolution.

3.3.2. GAT

To address the limitation that GNNs do not distinguish the different importance of neighboring nodes during aggregation, graph attention network (GAT) introduces the masked self-attention mechanism inspired by Transformers. When representing each node in the calculation graph, different weights are assigned to it according to the different characteristics of the neighboring nodes. GAT essentially has two kinds of operations: 1) Masked graph attention, that is, the operation of the attention mechanism is only performed on the neighboring vertices, which is also the commonly used way at present. 2) Global graph attention, that is, each vertex performs the attention operation on any vertex in the graph. The detailed calculation process of GAT is omitted here for brevity. Zhao et al. [23] used a GAT to explore the spatial correlation between wind variables for a graph constructed based on the MIC, which allows for the consideration of the combined effects of multiple variables on wind output. Specifically, they proposed a novel temporal attention layer to capture the long-term and short-term dependencies of the time series data of each input variable. The time steps that contribute more to future wind power generation are determined by calculating the attention coefficient at each step.

3.3.3. ACGRN

Adaptive graph convolutional recurrent neural network (AGCRN) does not have a complex network architecture and does not rely on a predefined graph structure to capture spatial sharing patterns. It has the advantages of being efficient and effective in graph neural network models, and has low computational time cost and excellent prediction performance [88]. AGCRN introduces two adaptive modules to enhance graph convolution: the node adaptive parameter learning module and the data adaptive graph generation module, which enable AGCRN to learn node-specific patterns without predefined graphs. In addition to modeling spatial relationships, AGCRN also combines gated recurrent units (GRUs) to track temporal dynamics. This approach integrates NAPL-GCN, DAGG, and GRU to encapsulate node-specific spatiotemporal patterns into sequences [25]. Liu et al. [70] proposed a hybrid DL model based on spatio-temporal graph contrastive learning. This model combines AGCRN with LSTM to further establish the connection between representations and output, thus effectively capturing the long-term dependencies in wind power sequences. Based on graph theory, Wang et al. [25] used spectral clustering to optimally partition the graph, thereby reducing computational cost and improving prediction accuracy. They then introduced AGCRN to capture the complex spatiotemporal correlations between wind turbines in each cluster and achieved satisfactory ultra-short-term clustering predictions by stacking two layers of networks.

3.3.4. STGNN

Spatio-Temporal GNN adds the dimension of time to the original graph, which means that the node features of the graph will change over time. To capture spatio-temporal dependencies among wind farms, Liu et al. [89] modeled wind farm clusters as a dynamic spatio-temporal graph. Instead of training the spatiotemporal GCN model on all data directly, they incorporated power fluctuation pattern partitions (PFPA) to enable fine-grained forecasting, training a separate STGCN model for each PFPA category. For large-scale wind farm cluster forecasting, Wang et al. [90] proposed an adaptive spatiotemporal graph convolution framework, in which multiple wind farms were represented as graph nodes and their spatial correlations were modeled through a dynamically adjusted graph structure. By combining improved multi-head attention, Chebyshev graph convolution, multi-scale temporal convolution, and multi-task learning, the model was able to synchronously generate multi-step forecasts for multiple wind farms and showed strong ultra-short-term forecasting performance. Compared with LSTM models that disregard spatial correlations or traditional machine learning approaches such as XGBoost, STGCN leverages graph convolution to effectively integrate the dynamic spatiotemporal relationships among wind farms, thereby reducing forecasting lag effects. More recently, spatiotemporal graph-based forecasting has continued to evolve toward more unified modeling of spatial and temporal dependencies. For example, Aquino et al. [91] proposed a StemGNN-based framework enhanced with a deep equilibrium model (DEQ), in which GNNs were used to represent turbine interactions while Seq2Seq and spectral-domain learning were employed to capture temporal and inter-series dependencies, further demonstrating the potential of graph-based spatiotemporal forecasting in WPF.

3.4. RNN-based models

3.4.1. LSTM

The structure of the memory cell in LSTM is shown in Fig. 7(c) and its calculation process can be expressed as follows:

$$\begin{aligned} i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i), \\ f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f), \\ g_t &= \tanh(W_g \cdot [h_{t-1}, x_t] + b_g), \\ o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o), \\ c_t &= f_t \odot c_{t-1} + i_t \odot g_t, \\ h_t &= o_t \odot \tanh(c_t), \end{aligned} \tag{5}$$

where x_t denotes the input vector at time step t, h_{t-1} and h_t denote the hidden states at time steps $t-1$ and t, respectively, c_{t-1} and c_t denote the cell states at time steps $t-1$ and t, respectively, i_t, f_t, and o_t denote the input gate, forget gate, and output gate, respectively, g_t denotes the candidate cell state, W_i, W_f, W_g, and W_o are the corresponding weight matrices, b_i, b_f, b_g, and b_o are the corresponding bias vectors, $[h_{t-1}, x_t]$ denotes the concatenation of h_{t-1} and x_t, $\sigma(\cdot)$ denotes the sigmoid activation function, $\tanh(\cdot)$ denotes the hyperbolic tangent activation function, and $\odot$ denotes element-wise multiplication.

LSTM models are highly dependent on hyperparameter selection, while traditional manual tuning is inefficient and prone to local optima. In the COA-CNN-LSTM hybrid DL model proposed by Abou Houran et al. [92], the Coati Optimization Algorithm (COA) is employed to automatically search for the optimal combination of LSTM hyperparameters. The CNN module extracts local spatial features from the input data, while the LSTM further learns the temporal evolution patterns of these features. Their integration achieves a “spatial feature extraction + temporal dynamics modeling” effect. Liu et al. [93] proposed the MTTFA-LSTM model, and the multi-step WPF task was reformulated as a multi-task learning problem. Specifically, all tasks share the global features extracted by the LSTM in the task-sharing layer. This design avoids the redundancy of training a separate model for each task, thereby reducing the number of model parameters and enhancing generalization capability. Mo et al. [94] proposed the FDNet model, which adopts a dual-LSTM architecture comprising an intra-patch LSTM and an inter-patch LSTM. The two modules focus on local and global temporal patterns, respectively, addressing the limitation of conventional LSTM in capturing multi-scale temporal dynamics. Furthermore, by leveraging patching operations and parameter sharing, the model reduces the number of LSTM iterations and parameter size, mitigating computational

redundancy when processing long sequences. To address the complexity and high computational cost of standard LSTM architectures, Wang et al. [33] proposed the Dual Layer–Shared Weight LSTM. Unlike the standard LSTM, which controls information flow through separate input, forget, and output gates, SWLSTM merges these into a single shared gate, reducing both model parameters and computational overhead while preserving the ability to capture long-term dependencies in time series, thereby enabling faster convergence. By employing a dual-layer structure to separately handle power forecasting and error compensation, and optimizing the input through data preprocessing, the model achieves high-accuracy and high-efficiency forecasting.

By introducing an additional backward input layer, BiLSTM can simultaneously exploit forward and backward information from historical data, thereby capturing bidirectional temporal dependencies in long sequences more comprehensively. This design effectively alleviates the vanishing gradient problem in conventional LSTM when processing long sequences, while enhancing the model's ability to retain latent patterns in historical data. In the personalized robust federated learning framework proposed by Zhao et al. [95], BiLSTM serves as the core predictive model for local training. Ye et al. [26] used BiLSTM as the encoder of the model, providing a rich context-aware feature foundation for subsequent decoder-based predictions. In Ref. [96], BiLSTM is used as a core component in the unary potential function of the Continuous Conditional Random Field (CCRF) to capture the nonlinear mapping between input features and wind power. This approach retains the nonlinear fitting capability of DL while leveraging the CCRF's strength in modeling internal dependencies within sequences. Experimental results indicate that, across all four seasons and various forecasting horizons, the BiLSTM model consistently achieves lower errors than other models, with particularly stable performance during summer and autumn when meteorological conditions are more complex. In the YJQR-LSTM model proposed by Wang et al. [32], the LSTM gating structure captures the dynamic characteristics of wind power data and generates hidden states rich in temporal information. These hidden states are then mapped via a fully connected layer to the core parameters of YJQR, providing the basis for probabilistic density forecasting. In the CNN-LSTM-AM model proposed by Sun et al. [66], the key parameters of the BiLSTM were fine-tuned through experimentation, with results showing that the model achieved optimal performance when the number of units was set to 64. Furthermore, comparative experiments demonstrated that the multivariate BiLSTM-Multi outperformed the univariate BiLSTM, confirming the effectiveness of incorporating multivariate features in improving WPF accuracy. Liu et al. [73] employed a modified encoder–decoder (MED) model based on LSTM, in which both the encoder and decoder are composed of LSTM units. In the experimental setup, a BiLSTM model was used as a baseline to verify the effectiveness of the unidirectional LSTM when combined with the Temporal Lag Attention (TLA) mechanism. While BiLSTM can capture both forward and backward dependencies in historical data, experimental results showed that the MED model with unidirectional LSTM outperformed its bidirectional counterpart when integrated with the TLA mechanism. This is because the TLA mechanism already corrects the temporal bias in numerical weather prediction (NWP) data through lag adjustment; thus, the bidirectional structure may introduce redundant information and increase complexity, resulting in negligible accuracy gains.

3.4.2. GRU

The gated recurrent unit (GRU) is a variant of RNN. The structure of the GRU cell is shown in Fig. 7(c) and the calculation formula is as follows:

$$\begin{aligned} z_t &= \sigma(W_z \cdot [h_{t-1}, x_t]), \\ r_t &= \sigma(W_r \cdot [h_{t-1}, x_t]), \\ \hat{h}_t &= \tanh(W_h \cdot [r_t \odot h_{t-1}, x_t]), \\ h_t &= (1 - z_t) \odot h_{t-1} + z_t \odot \hat{h}_t, \end{aligned} \tag{6}$$

where x_t denotes the input vector at time step t, h_{t-1} and h_t denote the hidden states at time steps $t-1$ and t, respectively, z_t and r_t denote the update gate and reset gate, respectively, $\hat{h}_t$ denotes the candidate hidden state, W_z, W_r, and W_h are the corresponding weight matrices, $[h_{t-1}, x_t]$ denotes the concatenation of h_{t-1} and x_t, $\sigma(\cdot)$ denotes the sigmoid activation function, $\tanh(\cdot)$ denotes the hyperbolic tangent activation function, and $\odot$ denotes element-wise multiplication.

Similar to LSTM, it was proposed to address challenges such as long-term dependency and gradient-related issues in backpropagation. Unlike LSTM, GRU has a simpler architecture, containing only two gating mechanisms: the reset gate and update gate, which result in higher computational efficiency, making it more suitable for real-time WPF tasks. To accommodate the temporal characteristics of wind power data, Zhao et al. [30] proposed a hybrid VMD-CNN-GRU model, in which the GRU network architecture and key parameters were specifically tailored, with the optimal configuration determined to be a four-layer GRU. By synergistically integrating noise reduction via VMD and spatial feature extraction through CNN, the model ultimately achieved high accuracy and stability in short-term WPF. Mudassar et al. [97] proposed GWO-DAGRU, a hybrid short-term WPF framework that combines GWO-based feature selection with a double-attention GRU predictor, improving the model's ability to capture informative inputs and temporal dependencies. Qin et al. [98] proposed an MSWT-MAMG framework that combines multi-scale wavelet transform, a multi-view attention mechanism, and dual recurrent neural networks composed of MGU and GRU. By introducing multi-view attention into the update gates of the two recurrent units, the model enables mutual information exchange between MGU and GRU, thereby improving temporal feature extraction and short-term forecasting accuracy.

In the Stacking Deep Learning Model (SDLM) proposed by Xing et al. [31], multimodal data are first processed through low-rank matrix fusion (LMF) to obtain a unified feature representation matrix. Subsequently, BiGRU and a leaky echo state network (LESN) jointly form the first-layer predictive submodels of the SDLM. Their prediction outputs are fed into a multiple linear regression (MLR) meta-learner, which performs weighted fusion of the strengths of the two submodels to produce the final prediction. This architecture leverages BiGRU's capability to capture bidirectional temporal dependencies and the LESN's advantages in model width and training efficiency, achieving structural complementarity and thereby enhancing the model's robustness against data uncertainty. In the MLRF–NNClustering–WOA–BiGRU model proposed by Jiang et al. [99], the multi-label ensemble random forest (MLRF) performs feature selection for multi-step forecasting tasks, identifying the optimal input features and time lags to reduce redundancy, lower computational cost, and improve generalization capability. The neural network clustering (NNClustering) module partitions the data into multiple clusters based on patterns, with each cluster modeled separately to reduce modeling complexity and enhance forecasting accuracy. For each clustered subset, a WOA–BiGRU model is constructed, where WOA is employed to automatically tune the key hyperparameters of BiGRU, thereby yielding the optimal forecasting model.

3.5. Comparative analysis

Table 3 compares several representative deep representation models for WPF from the perspectives of suitable application scenarios, input data, prediction horizons, advantages, and limitations. The comparison results show significant differences in modeling focus and practical applicability among different model architectures. RNN-based models are particularly effective at capturing nonlinear temporal dynamics from sequential data such as SCADA measurements, and they remain competitive for ultra-short-term to medium-term forecasting tasks. Owing to their strong temporal dependency modeling capability, they are generally more suitable for scenarios involving single turbines or individual wind farms. However, their ability to explicitly model spatial relationships is relatively limited. When spatial interactions among

Table 3
Comparison of representative deep representation models.

Model	Ref.	Suitable application scenarios	Input data	Prediction horizon	Advantages	Limitations
RNN-based	[27,30,32,33,92,93,97–99]	Single turbine; Wind farm	SCADA; NWP	Ultra-short-term; Short-term; Medium-term	Strong temporal dependency modeling; effective for nonlinear dynamic sequences	Limited spatial modeling ability; Sequential training is less efficient
GNN-based	[23,25,68–70,87–90]	Wind farm; Wind farm cluster	SCADA; NWP; Spatial relations	Ultra-short-term; Short-term; Medium-term	Explicitly captures spatial dependencies; Supports spatiotemporal joint modeling; Suitable for wind farm and wind farm cluster forecasting	Highly dependent on graph construction quality; Often computationally complex; Robustness to noisy and dynamic correlations still needs improvement
LLM-based	[48,65,67,72,78–82]	Wind farm; Wind farm cluster	SCADA; NWP; Multimodal data; Prior knowledge	Ultra-short-term; Short-term; Medium-term; Long-term	Leverages pre-trained knowledge; Supports prompt-based adaptation; Benefits from multimodal or exogenous information	Sensitive to prompt design and modality alignment; Adaptation frameworks can be complex; Interpretability remains limited

turbines or sites play an important role, GNN-based models offer a more suitable solution. By introducing graph structures, these models can explicitly characterize spatial dependencies and support joint spatiotemporal modeling, making them particularly advantageous for multi-turbine and cluster-level forecasting scenarios. Nevertheless, their performance is often highly dependent on the quality of graph construction, and the associated computational complexity may become a challenge in large-scale applications. LLM-based models, by contrast, provide a promising direction for more complex forecasting settings. These models can benefit from pre-trained knowledge and prompt-based adaptation, which gives them considerable potential for longer-term forecasting and broader applications in wind farm cluster forecasting. However, their practical effectiveness is still highly sensitive to model design, modality alignment, and adaptation strategies, while their interpretability remains limited. Encouragingly, the rapid development of LLM-related techniques may further improve their applicability in this field. Overall, model selection should be guided not only by forecasting accuracy, but also by application scenario, data availability, spatial scale, and computational cost. This suggests that the superiority of a model is inherently scenario-dependent rather than absolute. Therefore, a practically valuable comparison should focus not only on whether a method achieves lower prediction errors, but also on whether its modeling characteristics match the requirements of the target forecasting task.

4. Learning strategies

In addition to model architecture, learning strategies are also important for improving the adaptability of deep representation models in WPF. Among them, transfer learning and federated learning have been widely studied to address data scarcity, domain differences, and privacy-preserving collaborative forecasting [17].

4.1. Transfer learning

A large amount of data is required to train a deep representation network that can achieve accurate predictions. However, new wind farms often face a lack of historical operational data. The process of accumulating a dataset and training a model from scratch solely based on field data is often time-consuming and resource-intensive. A feasible research strategy to address this problem is to adopt a physics-based modeling approach. This approach aims to construct prediction models that do not rely on historical data by collecting operational state parameters of wind turbines in real time and incorporating environmental variables (e.g., temperature and geographic location). However, due to the significant differences in meteorological characteristics and operating conditions across different geographical regions, the models need to be highly customized for a specific site, which often leads to complex and computationally intensive model structures [100]. Another proven research strategy is the use of transfer learning. The core of this strategy is to first train a high-precision deep-representation prediction model using wind farms in the source domain with rich historical data, and then fine-tune the parameters of the pre-trained model based on the limited data in the target domain (newly built wind farms), so as to realize effective knowledge transfer and domain adaptation. In addition, transfer learning is not only applicable to newly established wind farms, but also to forecasting tasks for neighboring wind farms.

In WPF, wind power data are highly random and time-varying, and traditional models find it difficult to accurately capture its uncertainty. Tian et al. [29] adopted a strategy based on parameter sharing, which provides high-quality initialization for the target domain by sharing the optimal parameters of the source domain and accelerates model convergence. The target domain model can still generate high-precision prediction results when the amount of data is limited. However, simple parameter initialization or fixed-layer fine-tuning may lead to negative transfer when the source and target domains differ significantly. To address this, Jin et al. [101] proposed an Adaptive Transfer Learning (ATL) strategy. Instead of relying on manual experience to decide which layers to freeze, they employed evolutionary optimization algorithms to adaptively determine the optimal freezing strategy for the pre-trained model (specifically an LSTM with dual-stage attention). This ensures that only the general features applicable to the newly built wind farm are retained while domain-specific parameters are updated, effectively optimizing the model's adaptation structure. In order to solve the challenge of insufficient training data in privacy protection scenarios, Tang et al. [102] proposed a two-stage framework combining federated learning and transfer learning. To ensure data privacy, pre-training based on federated learning only transmits gradient updates and does not share raw data. In addition, its distributed learning does not significantly increase time consumption. Personalized transfer learning is then used to fine-tune the pre-trained model obtained through federated learning to obtain a personalized model suitable for the target wind farm. Differences in the geographical environments of wind farms may lead to poor performance of the global model at some sites. In order to solve the problem caused by the heterogeneity of data distribution of each wind farm in the federated learning scenario, Zhao et al. [95] proposed a personalized robust federated learning method. In the federated stage, the global model is first obtained through robust aggregation training, and then fine-tuned based on the limited data of the local wind farm, and some parameters are updated to obtain a personalized model. This approach retains the data privacy protection feature of federated learning, while solving the performance degradation problem caused by data heterogeneity, achieving higher prediction accuracy and better

generalization ability.Traditional transfer learning methods are mostly used to solve the spatial distribution differences between wind farms, but do not pay attention to the distribution changes of the same wind farm in different time periods. In view of the changes in the distribution, Tang et al. [74] regard historical time periods as different “source domains” and future time periods as “target domains” and divide the training data into several time domains. The data distribution in each time domain is relatively stable, while the distribution differences between time domains are as large as possible. The purpose is to construct a “worst case” so that the model faces the largest distribution difference between time domains during training, thereby improving the robustness when facing unknown distributions in the test phase. The model is encouraged to learn “time-domain-independent” features through adversarial training, and experimental results verify the effectiveness of this method. While the above methods focus on adapting model parameters to handle distribution shifts, transfer learning can also be utilized for feature augmentation to address the inherent information shortage in volatile wind sequences. Fan et al. [103] introduced a Dynamic Feature Transfer Learning (DFTL) framework, which treats high-similarity historical segments as source domains and the current prediction window as the target domain. By dynamically transferring periodic features from historical data to the current context through a hierarchical interactive network, this method effectively compensates for the lack of long-term periodic information caused by wind power randomness, improving prediction performance without requiring massive target datasets.

4.2. Federated learning

As mentioned above, training a prediction model that can extract deep representations requires a large amount of high-quality data. However, for commercial reasons and data security considerations, wind farms of different stakeholders often do not disclose private data, resulting in valuable wind farm data being in a “data island” state for a long time [104]. Therefore, it is of great significance to carry out research on cross-wind farm collaborative modeling under the background of ensuring the security and privacy of raw data. Federated learning [105] is the key technology that can solve this core problem. It allows multiple participants to collaboratively train a globally shared prediction model based on local private data, and achieves knowledge fusion only by exchanging model parameters, thereby fundamentally avoiding the direct transmission and centralized storage risks of sensitive data, and providing a feasible path to solve the contradiction between data security isolation and cross-domain collaborative modeling. The general structure and workflow of federated learning are shown in Fig. 8.

To solve data privacy and data island problems in WPF, Ahmadi et al. [106] adopted the federated learning framework, used LSTM as the base model for local training at each wind farm, learned the characteristics of wind power in its specific area, and aggregated the parameters of each local model through the FedAvg algorithm to generate a global model with strong generalization ability. Experiments show that this method avoids the risk of raw data leakage while maintaining high accuracy. Similarly, Li et al. [104] combined federated learning and deep reinforcement learning to propose the FedDRL framework. Each client uses deep deterministic policy gradient as the local prediction model, and the server uses the FedAvg algorithm to aggregate parameters. Experiments show that this framework achieves high forecasting performance under privacy protection, and only transmits model parameters instead of raw data, significantly reducing the communication load. However, the traditional FedAvg algorithm used in the above methods is sensitive to abnormal parameters and easily leads to the degradation of global model performance. To address this, Zhao et al. [95] proposed a personalized robust federated learning method, using the geometric median instead of FedAvg for parameter aggregation. By minimizing the sum of the distances from all parameter points to the aggregation point, the impact of outliers is reduced. In addition, considering that the data distribution of different wind farms is different and a single global model cannot adapt to all prediction tasks, they employed a two-stage strategy: after the federated learning training is completed, each wind farm uses a transfer learning strategy to fine-tune the global model to generate a personalized prediction model.

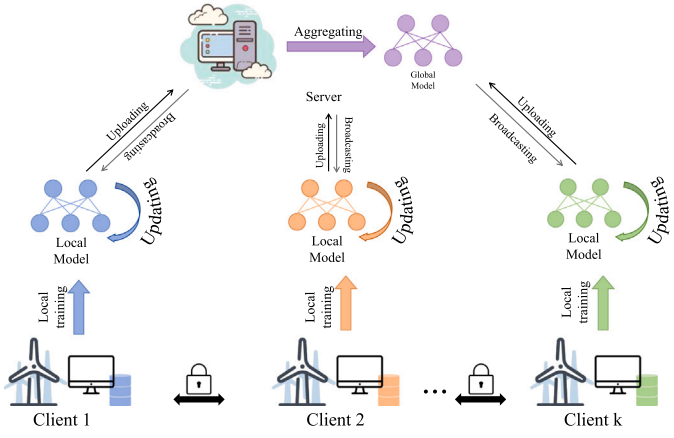


Fig. 8. Federated learning framework for privacy-preserving WPF.

While the two-stage strategy improves personalization, decoupling global pre-training from local fine-tuning may lead to information loss. To further optimize this, Han et al. [107] proposed a Federated Learning framework with Adaptive Parameter Updater (FLwAPU). Inspired by meta-learning, this method integrates the fine-tuning process directly into the federated iteration rounds. By adaptively updating local parameters based on the differences between global and local models, it enables simultaneous global optimization and local personalization, significantly improving performance in data-scarce wind farms compared to traditional decoupled approaches.

Beyond statistical heterogeneity, standard federated learning requires each client to train a complete model locally, which may be impractical for wind farms with limited computational resources or when integrating heterogeneous data sources. To address this computational challenge, Liu et al. [108] proposed a Federated Split Learning (FSL) framework. Unlike traditional federated learning that aggregates model parameters, FSL splits the neural network into server-side and client-side components. By exchanging only intermediate “smashed data” and gradients instead of full models, this approach not only preserves privacy through differential noise but also significantly reduces the computational burden on edge devices, offering a more flexible interaction paradigm for regional WPF.

Although both transfer learning and federated learning aim to improve the practical applicability of deep representation models, their roles in WPF are fundamentally different. Transfer learning mainly addresses the problem of insufficient labeled data and distribution mismatch by reusing knowledge from related source domains, and is therefore particularly suitable for newly built wind farms or cross-temporal adaptation tasks. However, its effectiveness largely depends on the similarity between source and target domains, and significant domain discrepancies may lead to negative transfer. By contrast, federated learning is designed for privacy-preserving collaborative modeling across multiple wind farms, making it more suitable for scenarios where raw data cannot be directly shared. Its main advantage lies in utilizing distributed data resources while preserving data security. Nevertheless, the performance of federated learning is often constrained by data heterogeneity, aggregation bias, and communication overhead, especially when local wind farms have highly imbalanced data distributions. In summary, transfer learning and federated learning are better understood as complementary rather than competing strategies. Emerging studies have combined the two to leverage both privacy-preserving collaboration and adaptive knowledge transfer [102].

5. Experiments

In this section, the experimental details of representative methods are systematically organized and compared, including the ratios of training, validation, and test splits, the prediction horizon, and performance across different evaluation metrics, as summarized in Table 4. Furthermore, we review how different studies set and justify the historical horizon, comparing their approaches in capturing temporal dependencies and balancing model complexity with forecasting accuracy, thereby highlighting the critical role of the historical horizon in WPF. In addition, the experimental contexts of inland and offshore wind farms are examined, taking into account wind resource characteristics, data availability, and environmental complexity, to elucidate the challenges and applicability differences faced by models across these scenarios. This section aims to provide a structured reference for future studies in experimental design and methodological selection.

Table 4 provides a structured summary of experimental settings used in representative deep representation learning-based WPF studies, enabling quick comparison across datasets, modeling choices, and evaluation metrics. This overview highlights the heterogeneity of experimental designs in current WPF research and underscores the need for standardized benchmarks.

5.1. Available datasets

In addition to forecasting methods, public datasets are essential for reproducible evaluation and fair comparison in WPF research. Recent studies have begun to systematically summarize publicly available wind power datasets and establish unified benchmark frameworks for cross-dataset evaluation, highlighting the importance of assessing model robustness under diverse geographical, operational, and data-distribution conditions [109,110]. Building on these efforts, we briefly summarize several representative public datasets that are widely used in the WPF literature for convenient reference.

SDWPF [39] This dataset is a spatial dynamic WPF dataset provided by Longyuan Power Group. It records operational data from 134 wind turbines in a wind farm every 10min over a period of 245 days. Each piece of data contains 13 characteristic dimensions that cover both the external environment that affects power generation, such as wind speed, wind direction, and ambient temperature, as well as key internal states, such as internal temperature. In addition, the SDWPF dataset contains the relative location coordinates of all turbines. It is worth noting that this dataset was used in the 2022 Baidu KDD Cup to evaluate the performance and limitations of WPF solutions at that time.

GEFCom2012 [111] This dataset was used for the wind power track of the Global Energy Forecasting Competition 2012. It records standardized wind power data and NWP data from seven wind farms for a period of nearly three years with a temporal resolution of one hour. Data from 2009-07-01 to 2010-12-31 are used as the training set, and data from 2011-01-01 to 2012-06-28 are used as the test set.

GEFCom2014 [112] This dataset was used for the wind power track of the Global Energy Forecasting Competition 2014. It records standardized wind power data and NWP data from 10 wind farms in Australia from 2012-01-01 to 2014-01-01 with a temporal resolution of 1h.

WIND [113] The Wind Integration National Dataset (WIND) Toolkit was released by the National Renewable Energy Laboratory (NREL) in 2013 with the aim of providing a comprehensive data platform for wind energy integration research. This dataset contains high-resolution wind resource data covering the contiguous United States, including multiple meteorological parameters such as wind speed, wind direction, temperature, and atmospheric pressure, spanning the period from 2007 to 2013.

Turkey [109] This dataset originates from the SCADA system of a wind turbine operating in Turkey and contains monitoring data recorded at 10-minute intervals during turbine operation. It includes variables such as the actual power output, wind speed and wind direction at hub height, as well as the theoretical power output corresponding to each wind speed as provided by the turbine manufacturer. These data enable the analysis of the relationships among wind speed, wind direction, and power generation, and can be further employed to assess and validate the accuracy of the turbine's theoretical power curve.

5.2. Evaluation metrics

In the WPF evaluation criteria, commonly used evaluation metrics for deterministic forecasting include the mean absolute error (MAE) [115], mean squared error (MSE) [24], root mean squared error (RMSE) [24], mean absolute percentage error (MAPE) [116], and the coefficient of determination (R^2) [117]. Since this review mainly focuses on deterministic WPF methods, probabilistic forecasting metrics, such as average coverage error (ACE) [118], prediction interval coverage probability (PICP) [119], and prediction interval normalized average width (PINAW) [120], are beyond the main scope of this paper. The definitions of the adopted metrics are given as follows:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{7}$$

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{8}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{9}$$

$$\mathrm{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{10}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{11}$$

where n denotes the total number of samples, i denotes the sample index, y_i denotes the actual value of the i-th sample, $\hat{y}_i$ denotes the corresponding predicted value, and $\bar{y}$ denotes the mean of the actual values in the test set.

Due to power differences among various wind turbines, relying solely on absolute error values makes cross-device and cross-scenario performance comparison difficult [121]. Therefore, some studies normalize the error by dividing the absolute error by the rated capacity P_{cap} of the device, converting it into a percentage error to achieve standardized evaluation. In addition to the commonly used metrics discussed in this section, other evaluation indicators frequently appear in the literature, including normalized mean absolute error (NMAE) [122], normalized root mean squared error (NRMSE) [122], and correlation coefficient (R) [123], which are also summarized in Table 4 for completeness.

5.3. Historical and prediction horizons

To date, there has been no universally accepted definition of the prediction horizon. Based on the temporal scale, forecasting ranges are commonly divided into four categories: ultra-short-term (a few seconds to 30min), short-term (30min to 6 hours), medium-term (6 hours to 1 day), and long-term (beyond 1 day) [14,15,124,125].

The historical horizon is a critical hyperparameter in forecasting models, as its setting requires a balance between model complexity and the sufficiency of predictive information [126]. A short historical horizon may provide inadequate information and lead to biased predictions, whereas an excessively long horizon increases computational burden and risks introducing noise and overfitting. This reflects the bias–variance trade-off, where the interaction between the historical horizon and model complexity governs the balance between generalization and fitting [3]. Hence, the selection of an appropriate horizon should be tailored to the characteristics of the specific model.

Zhao et al. [95] compared the prediction errors across different historical horizons {12, 18, 24, 30, 36} on the validation set and found that the error was minimized at 24, which was consequently adopted as

Table 4
Experimental details of representative methods.

Type	Ref.	Year	Model	Dataset	Ratio	Epochs	Horizon	Evaluation metrics	
								MAE/MAPE/NMAE/SMAPE/R/R^2	MSE/RMSE/NRMSE
LLM	[67]	2024	BERT4ST(110M)	Chuzhou(private)	7:1:2	70	10min/1h/4h	MAE(-) 1.20 /3.03 /6.78	RMSE(-) 2.19 /5.58 /11.91
	[83]	2026	Mistral(7B)	private	7:0:3	–	24h/96h/168h	MAE(24h) 7.3338/7.8297/7.5776/7.3491/7.6616/7.4643 R^2(24h) 0.8429/0.8240/0.8336/0.8423/0.8304/0.8379	RMSE(24h) 9.1350/9.8304/9.4771/9.1597/9.5943/9.3189
	[72]	2025	CMLLM-LLaMa(7B)	Dataset 1(private) Dataset 2(private) Dataset 3(private)	8:1:1	–	15m/1h/3h/ 6h/9h/12h	MAE(-) 0.3246/0.5688/0.7571/1.3221/1.5561/1.7121 MAE(-) 0.2394/0.4314/0.6553/1.0710/1.2762/1.3641 MAE(-) 0.3579/0.6714/0.7673/1.3950/1.4566/1.6653	RMSE(-) 0.4776/0.8868/1.1339/1.8753/2.1597/2.3493 RMSE(-) 0.4629/0.8622/1.2365/1.9317/2.1413/2.2311 RMSE(-) 0.5628/1.0402/1.1434/1.9704/2.0315/2.2791
	[80]	2025	WindPrompt Cast(58.3M)	public	8:0:2	–	4h	MAE(MW) 11.67	RMSE(MW) 18.82
	[65]	2025	GPT-2(124M)	private	–	10	168h	MAE(MW) 127 MAPE(%) 27.47 R^2 0.80	RMSE(MW) 213
	[79]	2025	GPT-2(82M)	Inner Mongolia(private) Gansu(private) Yunnan(private)	6:2:2	–	15m/1h/2h/4h	MAE(MW) 3.36/6.59/8.76/11.37 MAE(MW) 6.68/12.10/15.57/18.13 MAE(MW) 2.58/4.48/5.99/7.87	RMSE(MW) 5.24/9.57/12.21/15.26 RMSE(MW) 10.74/19.29/23.56/26.66 RMSE(MW) 4.64/7.40/9.27/11.59
GNN	[69]	2022	GCMCN	private	6:2:2	30	10m/20m/30m/ 40m/50m/1h/2h/3h	MAE(-) 0.0470/0.0622/0.0708/ 0.0735/0.0853/0.0908/0.1028/0.1193	RMSE(-) 0.0763/0.0989/0.1093/ 0.1167/0.1320/0.1375/0.1570/0.1718
	[87]	2023	AFS-OGS-GCN	private	8:1:1	–	1h/2h/3h/4h	MAE(%) 0.039/0.047/0.060/0.064	RMSE(%) 0.062/0.074/0.081/0.083
	[89]	2023	STGCN	private	8:1:1	–	1h/2h/4h	MAE(MW) 17.57/23.92/30.05	RMSE(%) 3.27/4.61/5.44
	[28]	2024	temporal–spatial GNN	SDWPF(public)	–	30	48h	MAE(-) 41.27	RMSE(-) 49.64
	[23]	2024	DA-STNet	SDWPF(public) NWWPF(private)	7:1:2	–	6h/12h/24h	MAE(MW) 1.122/1.819/2.469 MAE(MW) 0.179/0.233/0.275	MSE(MW) 0.296/0.589/0.876 MSE(MW) 0.070/0.106/0.136 RMSE(MW) 1.258/2.033/2.719 RMSE(MW) 0.205/0.267/0.320
	[68]	2025	STMGDDT	SDWPF(public)	8:1:1	50	10m/30m/1h/ 2h/3h/4h/5h/6h	NMAE(%) 1.8874/2.8369/3.4904/ 5.2989/6.8193/7.4545/8.9827/10.0052	NRMSE(%) 3.2485/4.5503/6.0697/ 8.2196/10.0911/11.5700/13.2085/14.6107
	[70]	2025	SGCL	SDWPF(public) HHL16(private)	6:2:2	300	1h/2h/3h	MAE(KW) 47.79/47.35/50.14 MAE(KW) 69.42/67.56/69.15	RMSE(KW) 75.42/75.74/80.10 RMSE(KW) 138.57/133.88/135.99
RNN	[33]	2024	DL-SWLSTM	private	8:0:2	80	15m/1h/2h	MAPE(%) Winter 1.15/2.41/6.95 MAPE(%) Spring 1.25/2.12/6.34 MAPE(%) Summer 2.01/4.12/7.14 MAPE(%) Autumn 1.45/2.47/5.96	RMSE(%) Winter 1.44/3.02/7.48 RMSE(%) Spring 1.62/3.34/7.53 RMSE(%) Summer 4.35/7.13/12.11 RMSE(%) Autumn 1.78/3.97/8.12
	[94]	2024	FDNet	GX dataset(private) HL dataset(private) LC dataset(private)	7:1:2	–	2h/4h/8h	MAE(-) 0.211/0.240/0.295 MAE(-) 0.309/0.430/0.560 MAE(-) 0.292/0.393/0.497	MSE(-) 0.152/0.188/0.244 MSE(-) 0.201/0.357/0.555 MSE(-) 0.193/0.319/0.507 RMSE(-) 0.390/0.434/0.494 RMSE(-) 0.448/0.597/0.745 RMSE(-) 0.439/0.565/0.712
	[93]	2024	MTTFA-LSTM	WIND(public)	8:0:2	–	1h/2h/3h	R^2 24h 0.8872/0.7733/0.6643 R^2 48h 0.8866/0.7727/0.6627 R^2 96h 0.8873/0.7742/0.6631	RMSE 24h 0.1120/0.1583/0.1929 RMSE 48h 0.1123/0.1586/0.1934 RMSE 96h 0.1120/0.1580/0.1933

(*continued on next page*)

Table 4 *(continued)*

Type	Ref.	Year	Model	Dataset	Ratio	Epochs	Horizon	Evaluation metrics	
								MAE/MAPE/NMAE/SMAPE/R/R^2	MSE/RMSE/NRMSE
	[26]	2024	TLA-MED	private	15:2:2	75	1h/2h/3h/4h	–	RMSE(%) Autumn 10.86%/14.28%/16.00%/16.74% RMSE(%) Winter 8.87% 11.94% 13.51% 14.55%
	[114]	2024	CEEMAN-EWT-LSTM	Turkey dataset(public) France dataset(public)	8:0:2	100	10m	MAE(KW) 51.61 MAPE(%) 1.43	RMSE(KW) 87.72
	[30]	2023	VMD-CNN-GRU	private	9:0:1	25	15m	MAE(MW) 0.8161 MAPE(%)11.62% R^2 0.9964	RMSE(MW)1.5651
	[31]	2023	MMF	inland(private) offshore(private)	8:0:2	–	1h/6h/12h/24h	MAE(-) Inland 0.0326/0.0335/0.0343/0.0361 MAE(-) offshore 0.0335/0.0343/0.0358/0.0364 SMAPE inland 0.1601/0.1621/0.1618/0.1636 SMAPE offshore 0.1727/0.1714/0.1721/0.1748 R inland 0.9709/0.9705/0.9702/0.9696 R offshore 0.9707 0.9702 0.9698 0.9685	RMSE(-) inland 0.0273/0.0286/0.0295/0.0318 RMSE(-) offshore 0.0285/0.0299/0.0311/0.0321
	[99]	2025	MLRF–NN Clustering–WOA–BiGRU	private	8:1:1	–	15m/1h	MAE(-) 2.6735/5.2482 R^2 0.9512/0.8324	RMSE(-) 4.2541/8.0458

the final input sequence length. Liu et al. [89] employed a grid search algorithm to validate hyperparameters by comparing prediction errors across different historical horizons, and reported that the optimal input windows for forecasting 1, 2, and 4 hours ahead were 2, 4, and 6 hours, respectively. Wang et al. [62], using the GEFCom2014 dataset, conducted a comparative analysis of different historical horizons {4, 8, 12, 24, 48, 72}. The results showed that 24 time steps yielded the best performance across all metrics, while 8 time steps also achieved satisfactory results in scenarios with limited data or strict real-time requirements, thus serving as a practical compromise. Liu et al. [63] calculated the PCC between input features and predicted power at different lags, showing that correlations decreased with increasing lag but remained above 0.8 for the first eight steps. Consequently, eight historical steps were selected as the input sequence length.

In terms of input sequence modeling, different strategies have been proposed. Mo et al. [94] employed a patching operation that divides a fixed-length historical window into equal-sized patches, extracting local temporal features within each patch and modeling long-term dependencies across patches. This approach alleviates the computational burden and gradient vanishing issues associated with directly modeling long sequences. Dai et al. [60], Liu et al. [89], and Hou et al. [68] adopted a sliding window strategy, where a fixed-length historical sequence is continuously updated and shifted forward in time to maintain temporal continuity in predictions. In contrast, Hu et al. [65] proposed a fixed-window approach that differs from the sliding window strategy by using a fixed-length historical sequence at each prediction step to output all future points simultaneously. This design avoids the overlap between inputs and outputs as well as the stepwise error propagation inherent in sliding windows, thereby effectively reducing the risk of data leakage.

5.4. Application analysis

Scenario analysis. WPF faces distinct challenges in inland and offshore application scenarios due to differences in terrain, atmospheric conditions, data characteristics, and operational requirements. For inland wind farms, abundant SCADA data and relatively extensive historical records usually provide a solid basis for developing data-driven forecasting models. However, inland wind farms are often influenced by complex terrain, vegetation, buildings, and local atmospheric effects, which introduce strong spatial heterogeneity and irregular disturbances into wind fields. As a result, wind speed and direction in inland areas often exhibit stronger fluctuations and nonstationarity, making the corresponding wind power series more random and uncertain, thereby increasing the difficulty of building stable and accurate forecasting models [3].

By contrast, offshore wind farms generally benefit from higher and more stable wind speeds, greater wind energy density, larger turbine capacities, and longer annual utilization hours, which make them highly attractive for large-scale renewable energy development. Nevertheless, the forecasting problem in offshore scenarios is far from simple. On the one hand, offshore wind power output is affected by complex and rapidly changing marine meteorological conditions, and the harsh marine environment is often accompanied by high construction and maintenance costs, limited long-term stable observations, and prolonged maintenance downtime, all of which may restrict data availability and increase forecasting uncertainty [127,128]. On the other hand, recent studies indicate that offshore WPF is also strongly influenced by wake interactions, wind-regime shifts, and pronounced multi-scale temporal dynamics. With the increasing density of offshore turbine deployment, wake effects and mutual interference among turbines become more significant, resulting in heterogeneous power behaviors across different turbine locations or regions within a wind farm [129]. In addition, offshore wind power series are often characterized by strong nonstationarity, spatiotemporal coupling, and multi-scale fluctuations, which make it difficult for a single global model to maintain robust performance under continuously changing operating regimes [130]. These characteristics explain why many recent offshore studies tend to adopt decomposition-based preprocessing, wind-regime clustering, attention mechanisms, or online/local learning strategies to better capture evolving marine wind patterns [131,132].

Moreover, offshore forecasting is more directly tied to short-term and ultra-short-term operational needs, such as real-time control, dispatch scheduling, and secure grid integration. Therefore, offshore models are usually required not only to achieve high statistical accuracy, but also to maintain robustness under rapidly changing marine weather conditions, wake-induced disturbances, and extreme events such as typhoons or strong offshore convection [133,134]. Overall, the main challenge of inland WPF lies in handling the uncertainty introduced by complex terrain and localized atmospheric disturbances, whereas offshore WPF must further address marine meteorological variability, wake effects, regime transitions, and stronger requirements for operational robustness.

Value-oriented analysis. Existing research primarily focuses on improving prediction accuracy through improvements in model architecture, feature extraction, and data processing strategies. However, the true purpose of the WPF task is to support the actual operation of power systems. Currently, most studies still evaluate forecasting models mainly from the perspective of statistical accuracy, whereas their operational value in dispatching and market decision-making remains relatively underexplored. Recent studies have pointed out that there is often a mismatch between the statistical value and the application value of WPF, because forecasts optimized only for homogeneous error metrics may fail to reflect the asymmetric operational risks caused by overestimation and underestimation in practical dispatch scenarios [135,136]. Therefore, when evaluating forecasting models, it is not sufficient to compare only model structures or error metrics [137]. Their usefulness for specific operational tasks, such as secure supply, renewable energy accommodation, dispatch planning, and reserve allocation, should also be taken into account.

In recent years, some studies have begun to move beyond purely accuracy-oriented evaluation and explicitly connect forecasting with downstream system and market operations. One important direction is value-oriented forecasting, which argues that the quality of forecasts should not be judged solely by statistical criteria, but also by the operational or economic value they create in subsequent decision-making [138]. In day-ahead and real-time operation, forecasts directly affect dispatch schedules, reserve procurement, and balancing actions. As a result, a forecast with smaller conventional errors does not necessarily yield lower operating costs or lower operational risks [139]. Based on this understanding, some studies formulate forecasting models with training objectives aligned to downstream decision value, such as minimizing overall operating costs in sequential dispatch problems. Some others extend this idea from point forecasting to uncertainty forecasting, showing that prediction intervals with better statistical properties do not always lead to better robust operational decisions [140]. In parallel, closed-loop predict-and-optimize frameworks further tighten the interaction between forecasting and optimization by feeding the results of downstream unit commitment, dispatch, or market-clearing problems back into the forecasting stage [136,141]. These studies suggest that forecasting models should be evaluated not only by how accurately they reproduce future power output, but also by how effectively they support economic, reliable, and risk-aware operational decisions under specific application scenarios.

These studies indicate that application-oriented evaluation should become an important complement to conventional error-based assessment in WPF. In particular, model comparison should consider whether a forecasting method can better support downstream operational objectives, rather than relying solely on MAE or RMSE [142]. From this perspective, the real benefits of different models should be understood in relation to the specific tasks they support. For example, models with stronger point forecasting capability may be more suitable for routine dispatch scheduling, whereas models with better uncertainty

Table 5
Summary of suitable forecasting models for different power market and industrial application scenarios.

Market and Application Scenario	Key Industrial Objectives	Forecasting Requirements	Suitable Representative Models
Day-ahead electricity market	Maximize bidding revenue; reduce market deviation risk	Medium-term forecasting; stable multi-step forecasting	RNN-based models [27,93]; Transformer/attention-based models [62,81]; decomposition-based hybrid models [44,58]
Real-time market; imbalance settlement	Support real-time trading adjustments; minimize real-time imbalance penalties	Ultra-short-term and short-term forecasting; high-frequency updating	Temporal attention models [60]; attention-based models [50,97,98]
Microgrid and energy management	Optimize energy storage dispatch; facilitate load matching; ensure operational stability and prevent blackouts in islanded mode	Ultra-short-term/short-term forecasting; uncertainty- and risk-aware forecasting	LLM-based models [82]; probabilistic or uncertainty-aware models [15,63,137]
Grid dispatch and reserve scheduling	Maintain supply–demand balance; dynamically allocate operational reserves; reduce dispatch costs and improve grid security	Ultra-short-term/short-term forecasting; reliable and uncertainty-aware forecasting	GNN-based models [89,90]; probabilistic or uncertainty-aware models [15,63,137]

representation may provide greater value for reserve allocation, robust operation, or market-oriented decision-making. Likewise, a model that does not achieve the lowest statistical error may still deliver better practical performance if it helps reduce costly underestimation events or better matches the needs of a particular decision problem. This perspective helps reveal the real benefits of different models in system and market operations, and also provides a more practical basis for model selection in different forecasting scenarios.

Market and industrial application analysis. Following the above discussion, application-oriented evaluation further requires a clearer understanding of how different forecasting models match the requirements of specific market and industrial scenarios. As emphasized in the authoritative report Innovation landscape for a renewable-powered future by the IRENA [143], advanced renewable forecasting is a foundational innovation for power system flexibility. The report points out that successful wind power integration strictly depends on aligning forecasting capabilities with diverse operational timescales, thereby imposing distinct forecasting requirements in terms of horizon, update frequency, stability, and uncertainty awareness across different application scenarios. Therefore, the suitability of a forecasting model should not be determined only by its overall prediction accuracy, but also by whether its modeling characteristics are consistent with the decision-making context in which the forecast is used. To provide a more application-oriented perspective, Table 5 summarizes the relationships among typical power market and industrial scenarios, their key operational objectives, forecasting requirements, and suitable model families.

Day-ahead electricity markets usually require medium-term and stable multi-step forecasting to support bidding strategies and reduce deviation risks. In this scenario, RNN-based models [27,93], Transformer/attention-based models [62,81], and decomposition-based hybrid models [44,58] are generally suitable, because they can capture temporal dependencies and improve the robustness of multi-step forecasting. For real-time markets and imbalance settlement, the forecasting task shifts toward ultra-short-term or short-term forecasting with high-frequency updating. The main operational objective is to support real-time trading adjustments and minimize imbalance penalties. Therefore, temporal attention models [60] and attention-enhanced RNNs [50,97,98] are more appropriate, as they can dynamically emphasize recent and informative historical patterns.

For microgrid and energy management applications, WPF is closely related to energy storage dispatch, load matching, and operational stability, especially under islanded or weak-grid conditions. These scenarios require not only short-term forecasting accuracy but also uncertainty- and risk-aware decision support. Accordingly, LLM-based models [82] and probabilistic or uncertainty-aware models [63,137] have potential advantages, especially when multi-source information, contextual knowledge, or few-shot adaptation is required. In grid dispatch and reserve scheduling, forecasts directly affect supply–demand balance, reserve allocation, dispatch cost, and system security. Compared with single-site forecasting, this scenario usually involves stronger spatial coupling among turbines, wind farms, and regional grids. Therefore, GNN-based models [89,90] are particularly suitable for capturing spatial dependencies and inter-node correlations, while probabilistic forecasting models [15,140,141] can further quantify forecasting uncertainty and support reliable reserve scheduling.

Overall, no single model family is universally optimal across all power market and industrial scenarios. Instead, model selection should be aligned with the operational objectives, forecasting horizons, spatial scale, and uncertainty requirements of each application. This observation further reinforces the need for future WPF studies to move beyond purely accuracy-oriented benchmarking toward value-oriented and scenario-aware evaluation.

6. Research challenges and future perspectives

6.1. Challenges and gaps

Data availability and reproducibility. At the data level, data quality and availability remain major bottlenecks in WPF. SCADA data often contain missing values, noise, and anomalies, while multi-source datasets may suffer from inconsistencies in temporal resolution, sampling frequency, and spatial granularity. In addition, differences in terrain, meteorological conditions, and farm-specific operating characteristics can cause substantial distribution shifts across wind farms, thereby limiting cross-site generalization. Another practical limitation is that many existing studies still rely on private industrial datasets, whereas high-quality publicly available datasets remain insufficient. Moreover, the lack of open-source code, preprocessing pipelines, and standardized evaluation settings weakens reproducibility and makes fair cross-study comparisons more difficult. Addressing these issues requires more emphasis not only on increasing data volume, but also on improving data usability, reproducibility, and cross-site collaboration. In particular, large-scale cross-regional benchmark datasets, standardized evaluation protocols, and open-source implementations are needed to support fair comparisons and more reproducible research.

Physics-informed deep representation learning. At the model level, a key research gap lies in the insufficient integration of physical mechanisms into deep representation learning frameworks. Although LLM-based, GNN-based, and RNN-based models have shown strong capability in capturing nonlinear temporal dependencies, spatial correlations, and multi-source feature interactions, most existing methods remain predominantly data-driven. They mainly learn statistical associations from historical SCADA, NWP, and wind power data, while wind energy conversion mechanisms, turbine operating characteristics, wake effects, and other physical constraints are not sufficiently incorporated

into the forecasting process. This limitation may weaken model robustness under extreme weather conditions, distribution shifts, and newly built wind farms with limited historical data. A promising direction is to further develop physics-informed deep representation learning. Instead of relying solely on data-driven feature extraction, physical knowledge can be introduced as prior information, model constraints, or domain-guided representations. For example, turbine power-curve characteristics, meteorological consistency, and spatial wake interactions may be incorporated into forecasting models to improve physical consistency and cross-scenario generalization. Such integration is expected to enhance not only forecasting accuracy, but also robustness and practical reliability in real-world wind power applications.

Model interpretability and trustworthy forecasting. In addition to the insufficient integration of physical mechanisms, model interpretability is another important challenge at the model level. Although advanced models such as LLMs, GNNs, RNNs, and attention-based architectures have achieved promising forecasting performance, their internal decision-making processes are often difficult to understand. In practical power system operations, forecasting results are not only used for accuracy comparison, but also provide support for dispatch scheduling, reserve allocation, and risk management. Therefore, a lack of interpretability may reduce the credibility of forecasting models and hinder their deployment in real-world decision-making scenarios. Further progress may be achieved by developing more interpretable and trustworthy forecasting frameworks. On the one hand, model-level interpretability can be improved by analyzing feature importance, temporal attention, spatial dependencies, and cross-modal interactions, thereby revealing how different meteorological variables, historical time steps, and wind farm nodes contribute to the final prediction. On the other hand, uncertainty-aware explanation methods should be incorporated to help operators understand not only the predicted value, but also the reliability and potential risks of the forecast. Enhancing interpretability is therefore essential for improving model transparency, operational trust, and practical applicability in WPF.

Value-oriented decision support. At the application level, a major challenge is that current WPF research still focuses predominantly on improving statistical accuracy, while the practical value of forecasting for power system operation remains insufficiently explored. In real-world applications, the ultimate purpose of WPF is not merely to reduce prediction errors, but to support downstream operational tasks such as dispatch scheduling, reserve allocation, renewable energy accommodation, and electricity market decision-making. However, a forecast with lower conventional error does not necessarily lead to better operational outcomes, because the statistical performance of a model may not fully coincide with its application value in real decision-making scenarios. In particular, forecasts optimized only for homogeneous error metrics may fail to reflect the asymmetric operational risks caused by overestimation and underestimation. This calls for a shift from accuracy-oriented evaluation toward application-oriented assessment. More specifically, forecasting models should be evaluated according to how effectively they support downstream objectives such as economic efficiency, operational reliability, and risk-aware decision-making. From this perspective, the benefits of different models should be understood not only in terms of predictive accuracy, but also in terms of the operational and economic value they create in specific application scenarios.

6.2. Future perspectives

After discussing the above challenges from data, model, and application perspectives, it is also worth noting that several emerging techniques may provide new opportunities for WPF. As discussed above, LLM-based forecasting represents one of the recent attempts to introduce emerging AI techniques into WPF. Beyond this line of research, time series foundation models (TSFMs) [84], physics-informed deep learning, especially physics-informed neural networks (PINNs) [144,145], as well as quantum computing-based forecasting, may further broaden the methodological landscape of WPF. TSFMs may enhance cross-domain adaptation and data-efficient learning through large-scale pretraining, while physics-informed methods can introduce physical knowledge into model training to improve physical consistency. Quantum computing-based forecasting also provides a new and potentially powerful computational paradigm. Through high-dimensional feature mapping and parallel processing, quantum-enhanced models may extract more effective representations with fewer parameters when dealing with complex and highly variable wind power forecasting scenarios [146–148]. Nevertheless, most of these techniques are still at an early stage in WPF, and their effectiveness, scalability, and practical deployment require further validation.

7. Conclusion

In recent years, deep representation learning for WPF has exhibited increasingly systematic, structured, and diversified development. With the growing availability of SCADA data, meteorological information, and wind farm layout data, research has shifted from traditional univariate time-series modeling to unified modeling of multi-source, multi-scale, and multi-dimensional dependencies. The modeling objective has likewise evolved from short univariate sequence forecasting to multi-source information fusion and the characterization of complex spatiotemporal interactions. Correspondingly, model architectures have rapidly progressed from conventional RNNs toward GNNs, Transformer-based frameworks, and LLM-based models. This evolution reflects the shift in models from capturing local temporal dependencies to achieving global spatiotemporal semantic understanding. Meanwhile, attention mechanisms have been widely applied across temporal, spatial, and feature dimensions, becoming a key component in improving model performance.

This review synthesizes advances in WPF from four perspectives—data, models, learning strategies, and experimental details. It summarizes the main approaches to SCADA data preprocessing and signal decomposition, outlines the characteristics of representative spatiotemporal modeling methods—including LLM-based models, GNNs, and RNNs—and further discusses the critical role of attention mechanisms as well as the use of transfer learning and federated learning for cross-farm collaboration. By integrating publicly available datasets, experimental configurations, and comparative analyses across different application scenarios, this review provides a systematic overview of the principal methods and development pathways of deep representation learning in WPF, offering a clear and coherent perspective on the technical evolution of the field.

CRediT authorship contribution statement

Decheng Liu: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Yunfei Yan:** Writing – original draft, Methodology. **Shuai Chen:** Writing – review & editing. **Bo Wang:** Writing – review & editing. **Xinghua Li:** Supervision. **Haibin Zhang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported in part by the National Natural Science Foundation of China under Grant 62571405, 62306227, 62125205, U23A20303; in part by the Natural Science Basic Research Plan in Shaanxi Province of China (Program No. 2025JC-JCQN-089); in part by Young Talent Fund of Association for Science and Technology in Shaanxi under Grant 20250143; in part by the Fundamental Research Funds for the Central Universities under Grant KYFZ26006.

Data availability

Data will be made available on request.

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